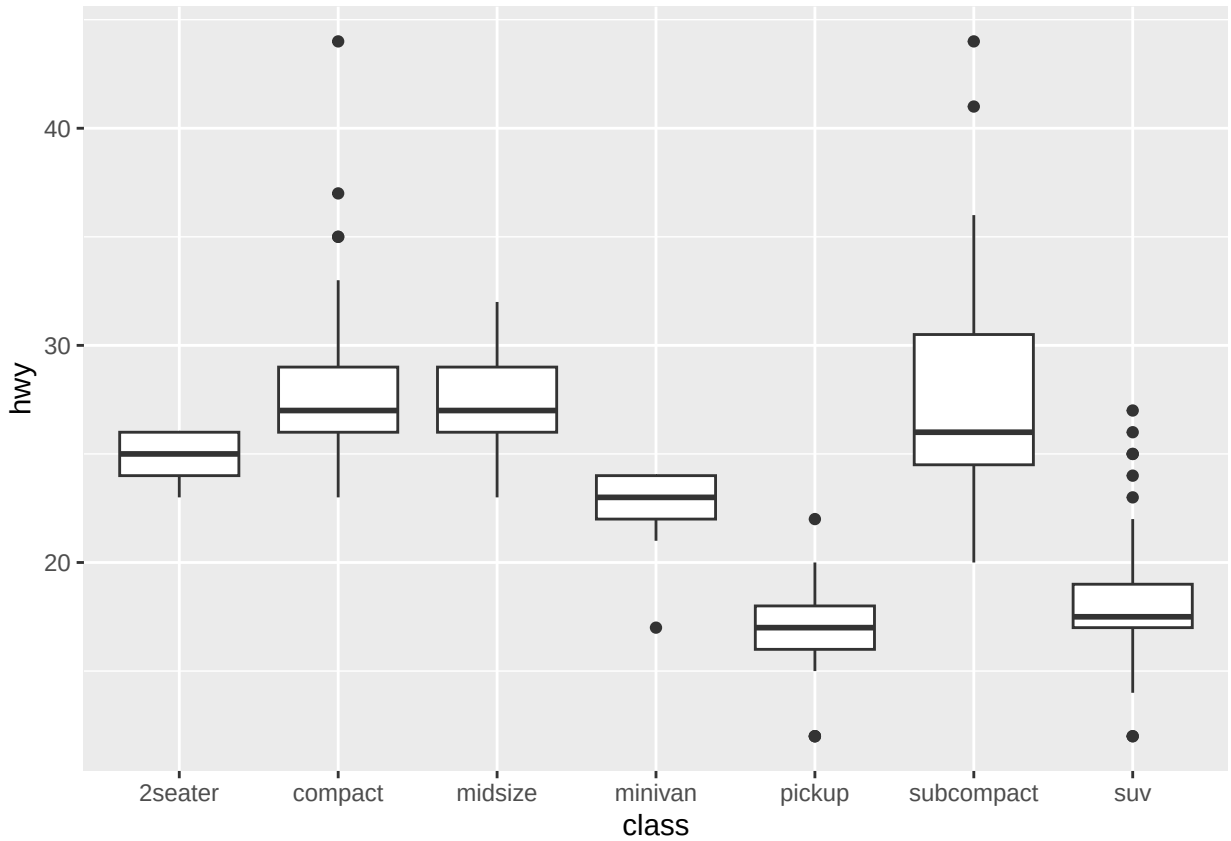
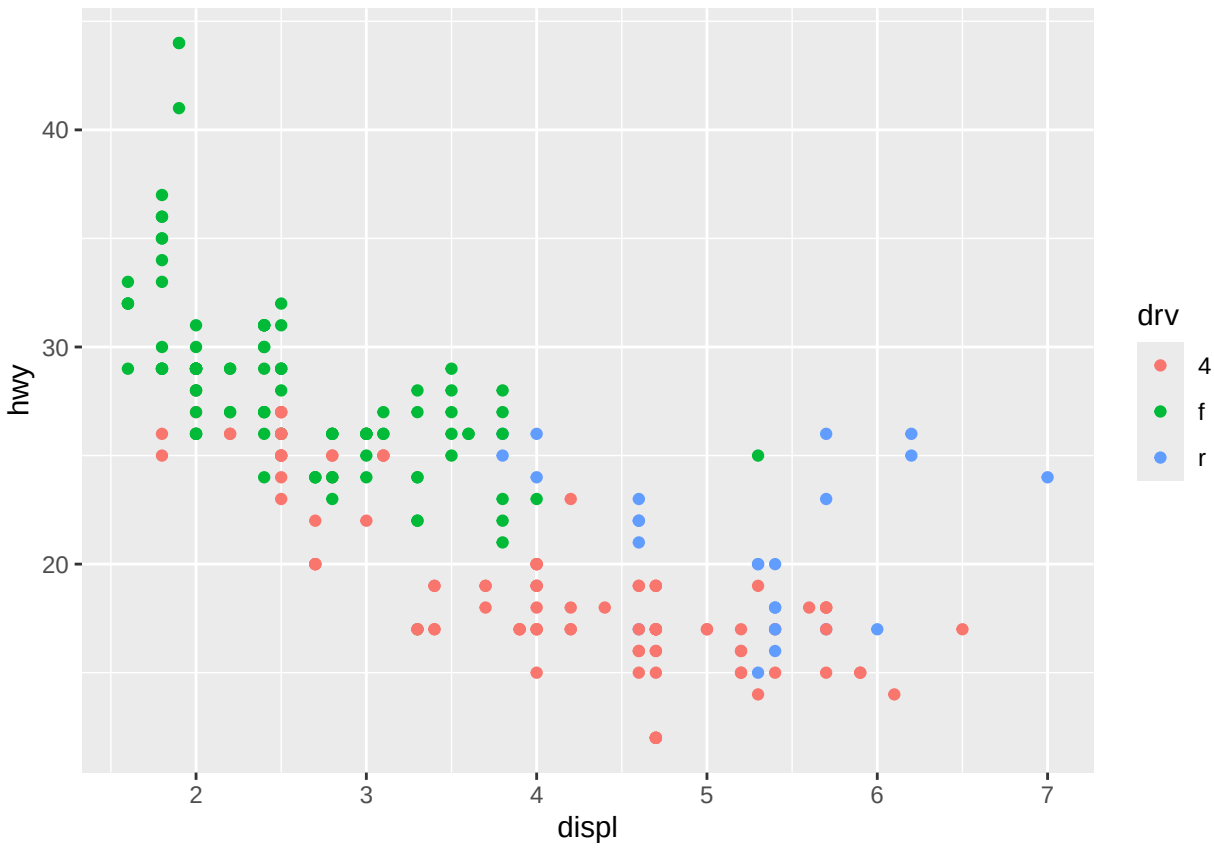
R Scribbles

Anurag (bmat2508@isibang.ac.in)

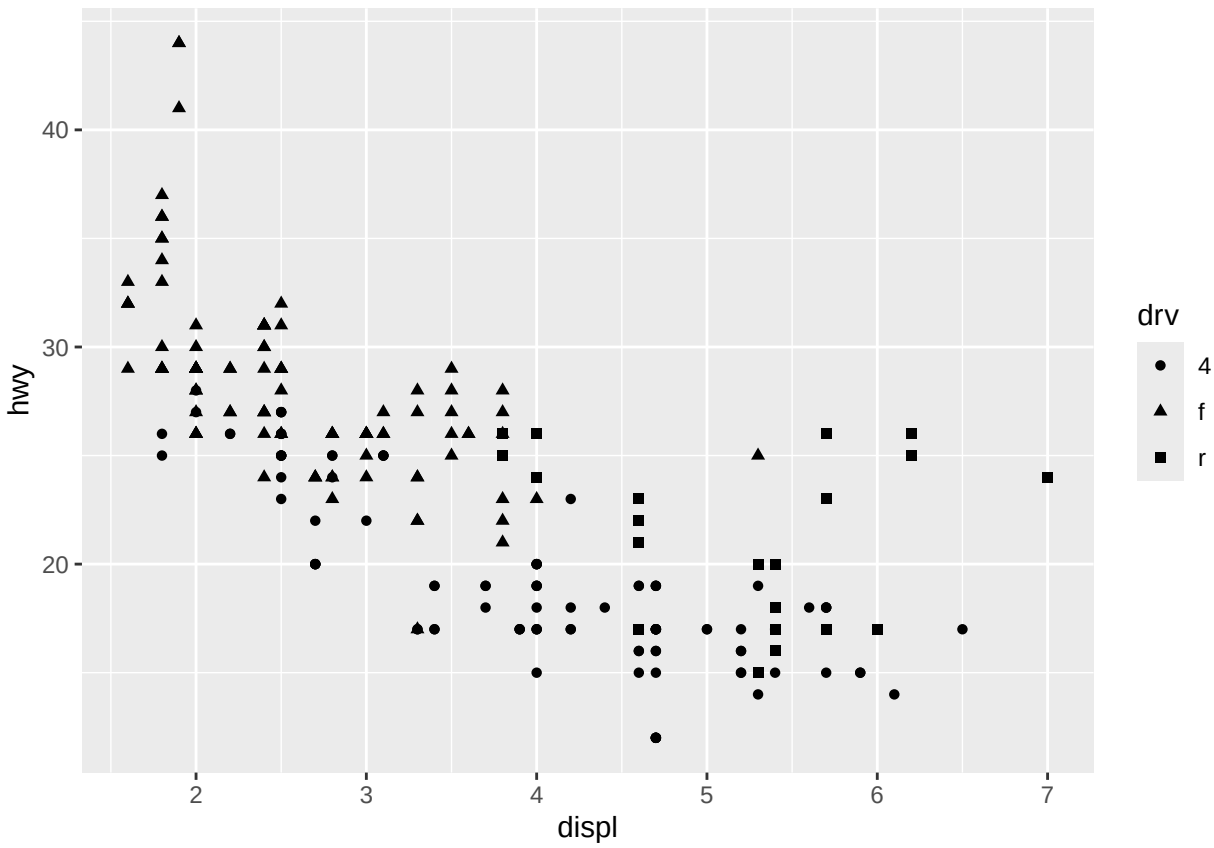
```r
ggplot(data=mpg,aes(x=class,y=hwy))+geom_boxplot()
```




```r
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))
```

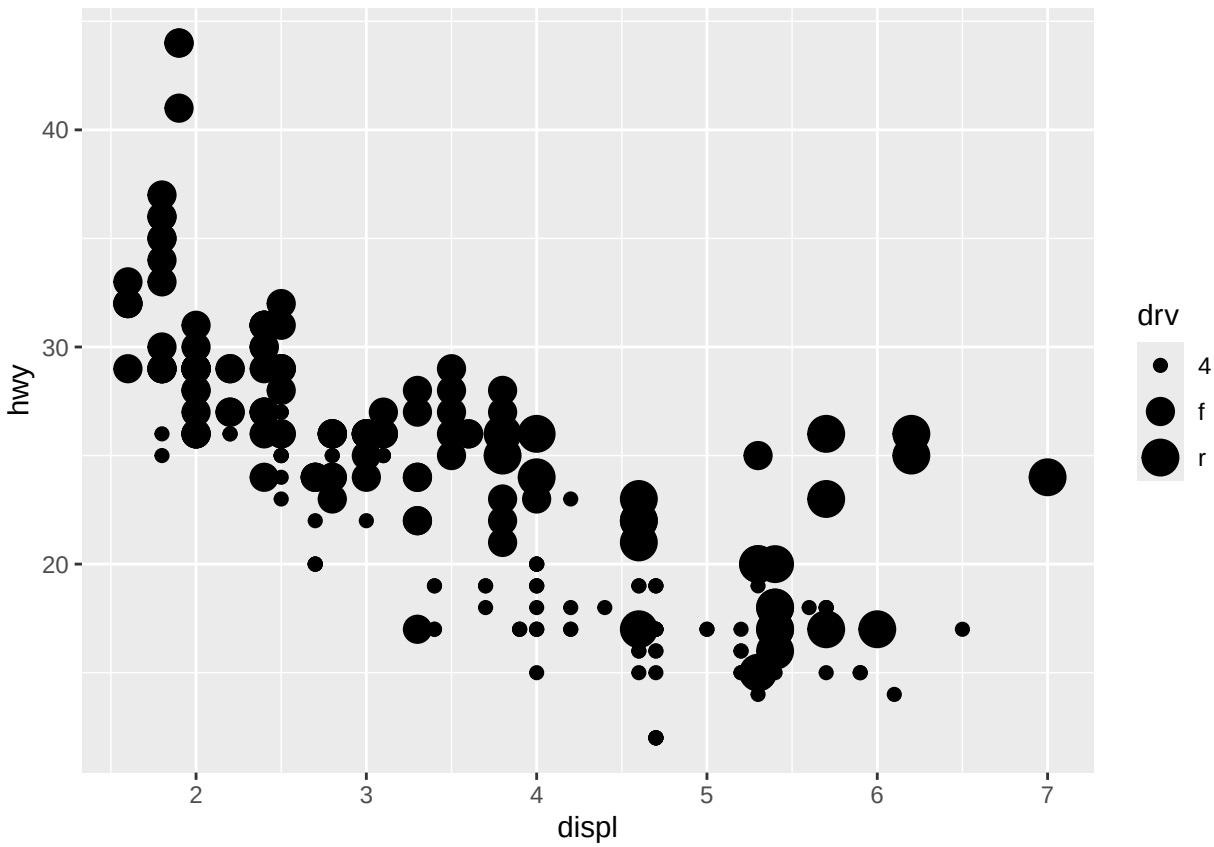



```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,shape=drv))
```



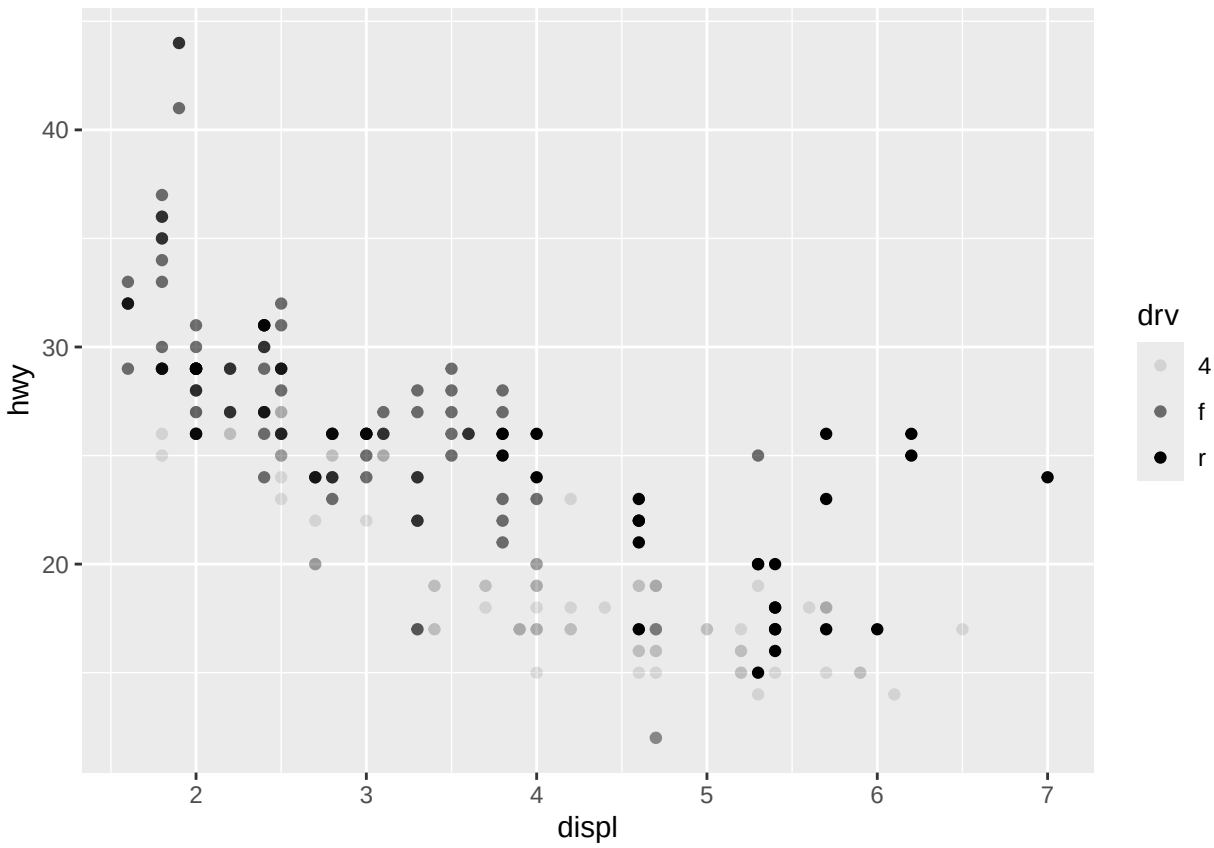

```r
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,size=drv))
```

```
## Warning: Using size for a discrete variable is not advised.
```

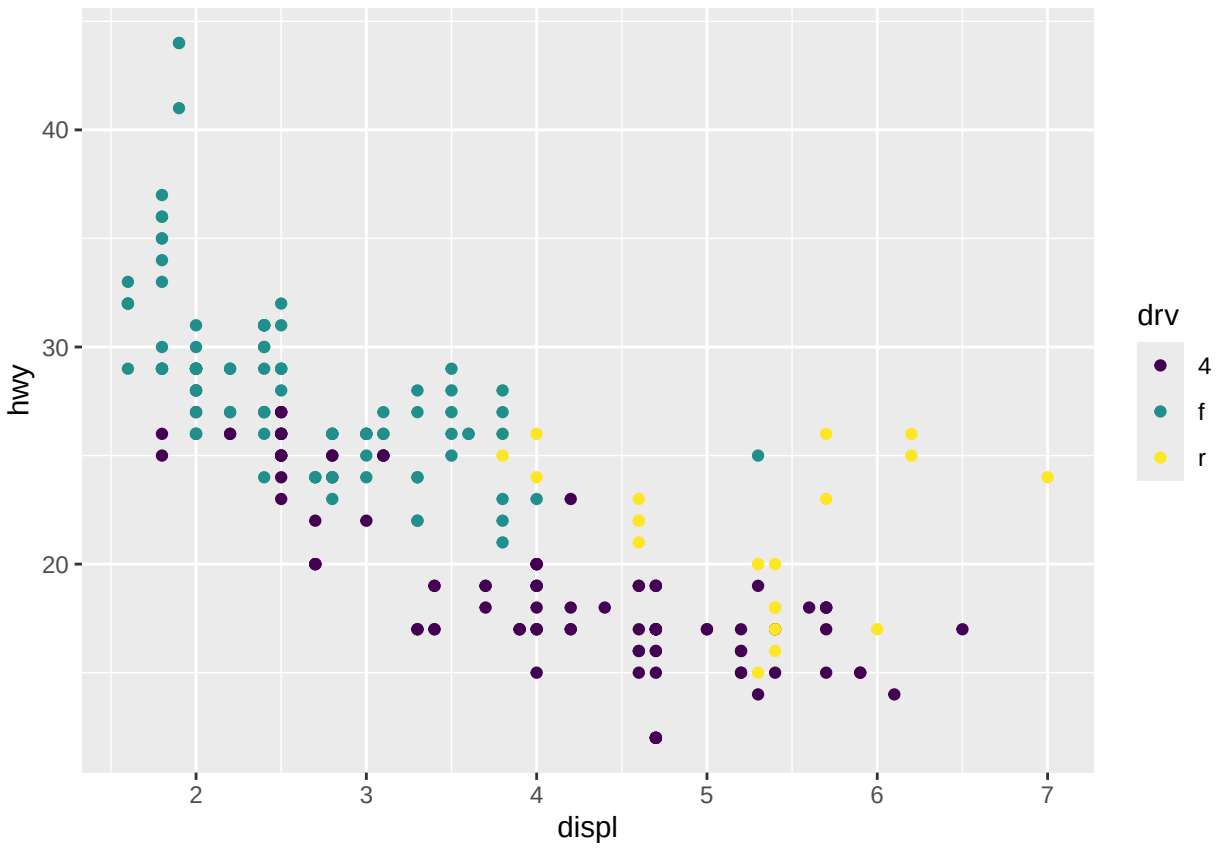



```r
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,alpha=drv))
```

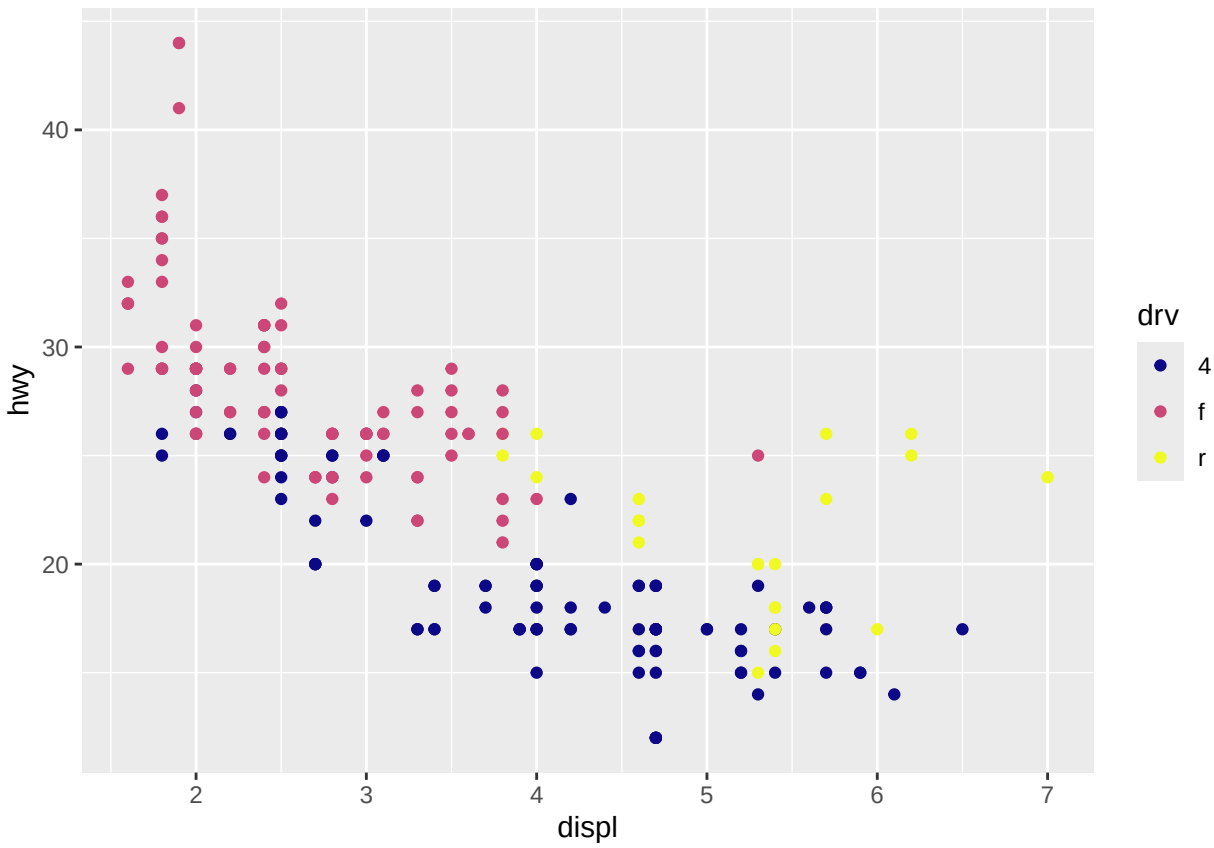
```
## Warning: Using alpha for a discrete variable is not advised.
```



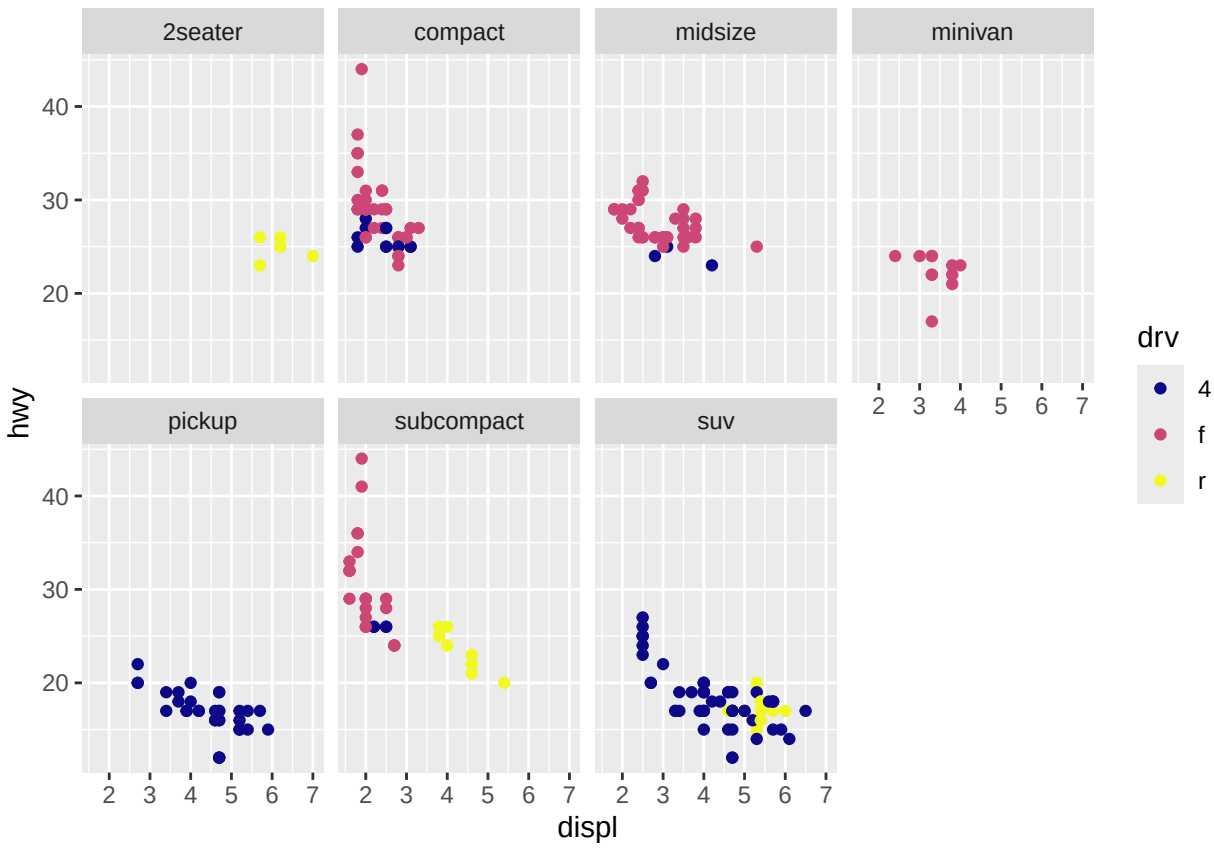

```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d()
```



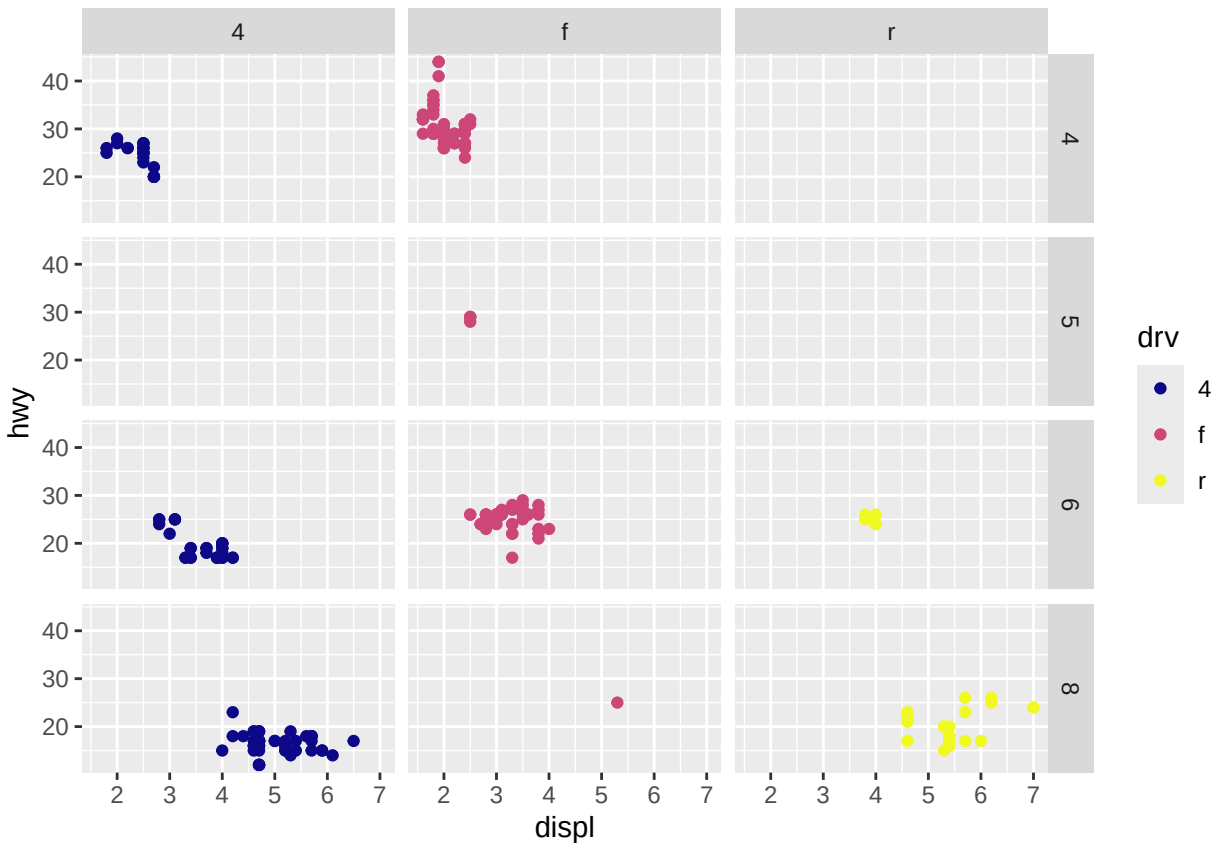

```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d(option="plasma")
```




```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d(option="plasma")+facet_wrap
```

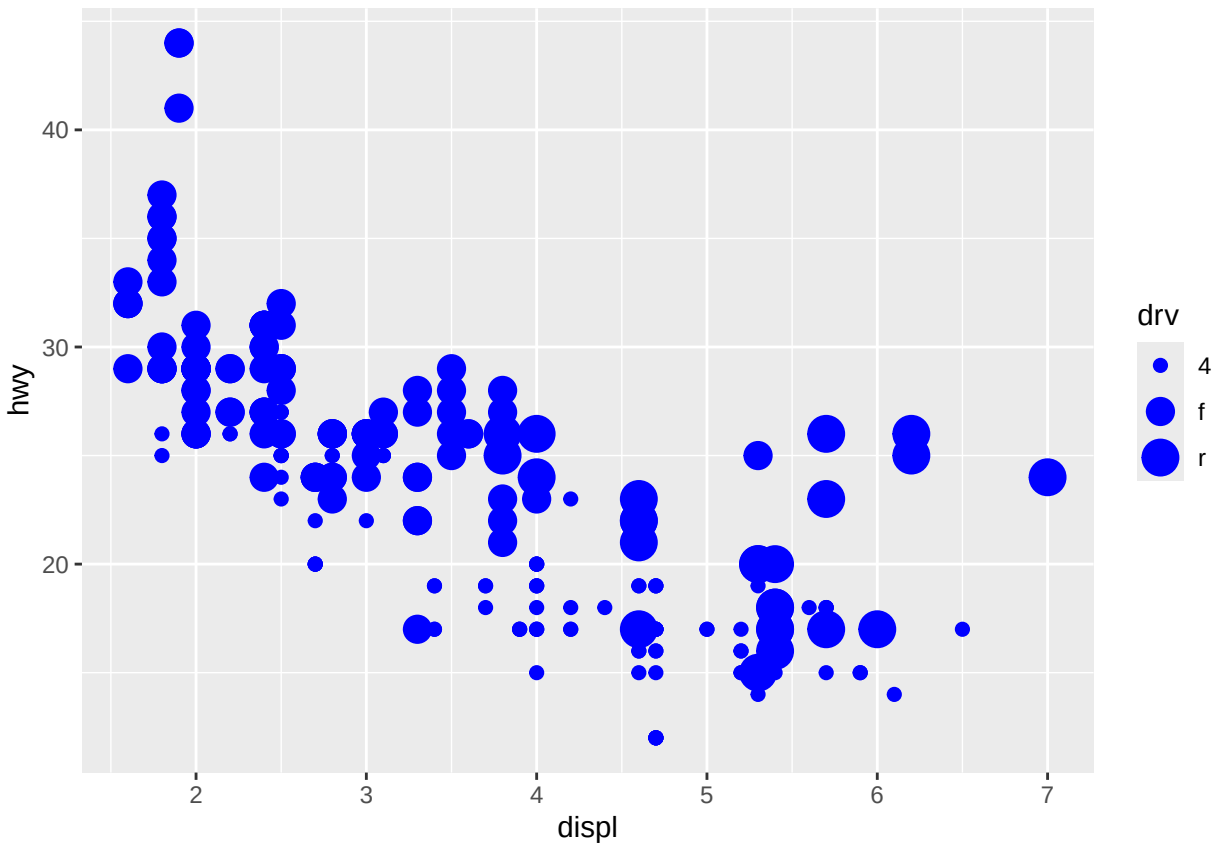



```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d(option="plasma")+facet_gri
```

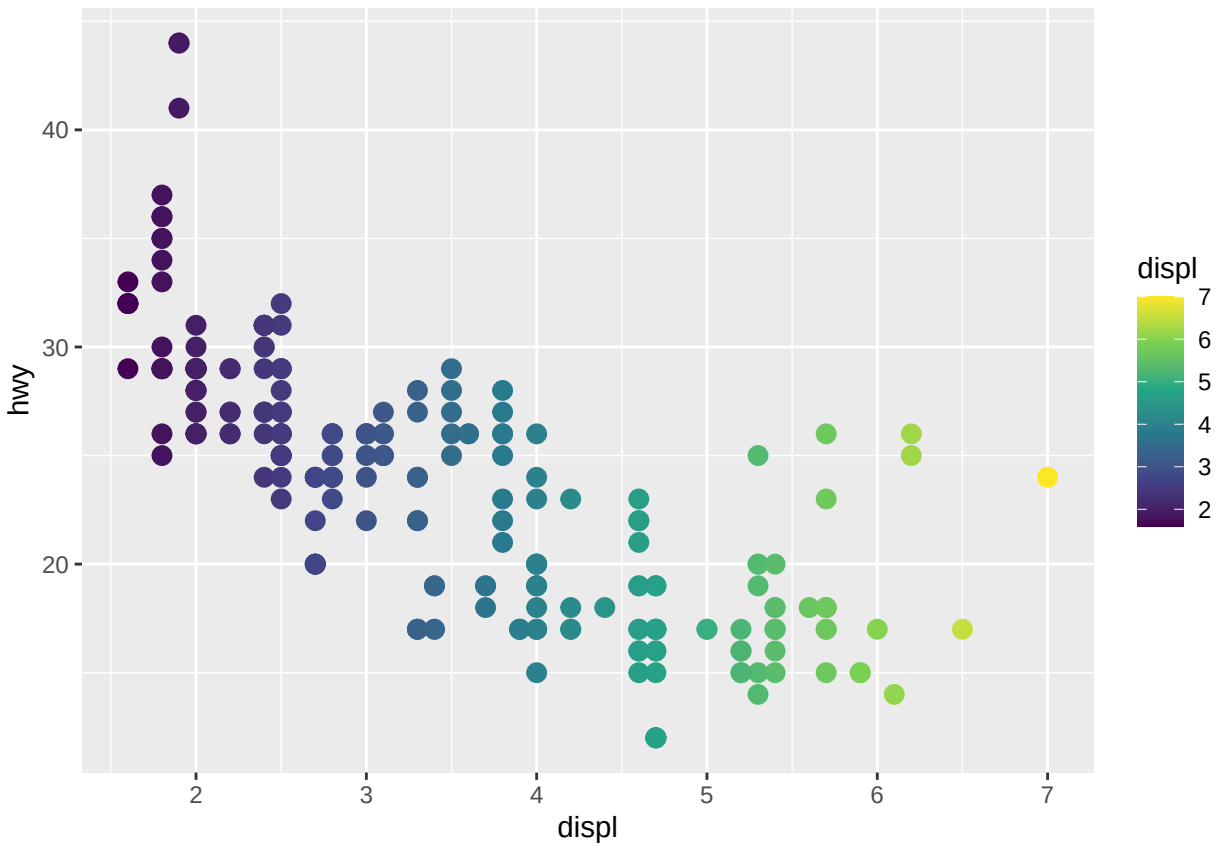



```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,size=drv),colour="blue")
```

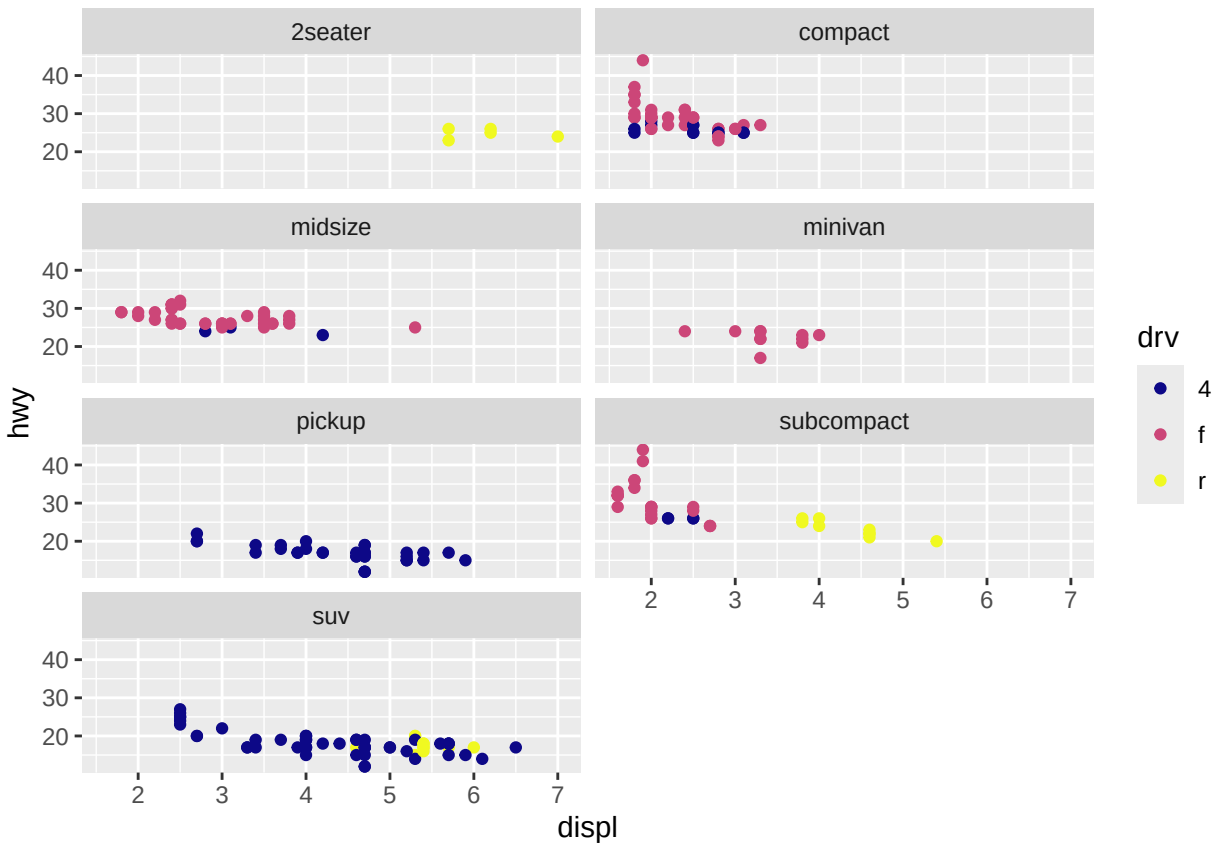
```
## Warning: Using size for a discrete variable is not advised.
```



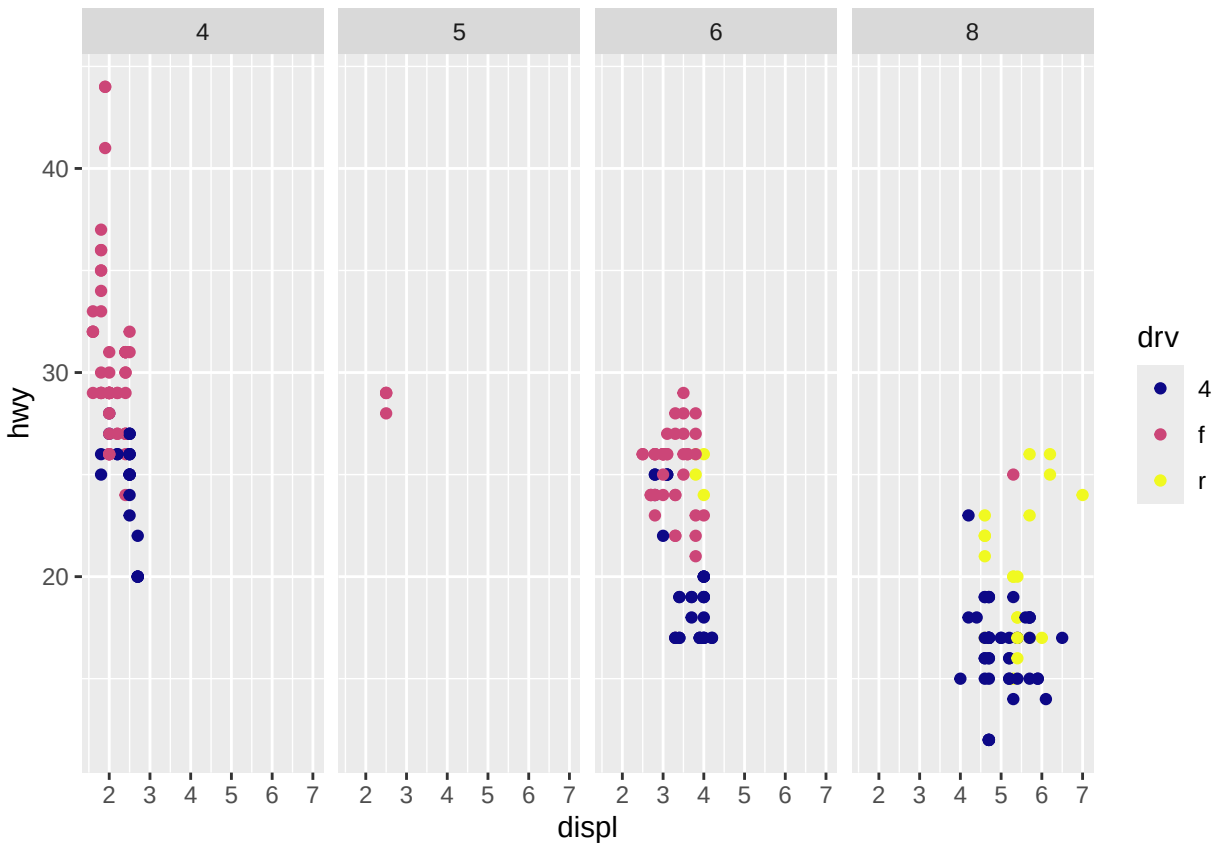

```
ggplot(mpg, aes(x=displ,y=hwy,colour=displ))+geom_point(size=3)+scale_colour_viridis_c()
```



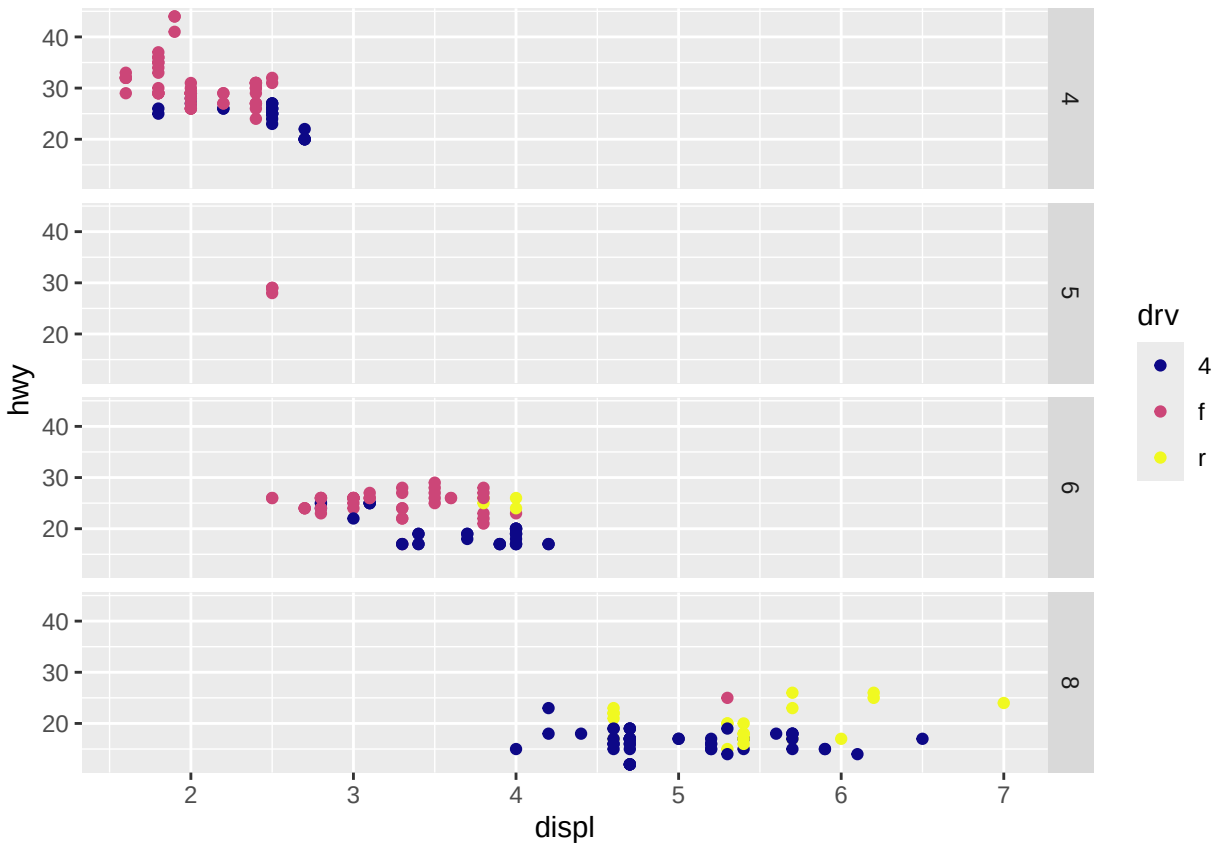

```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d(option="plasma")+facet_wrap
```




```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d(option="plasma")+facet_gri
```

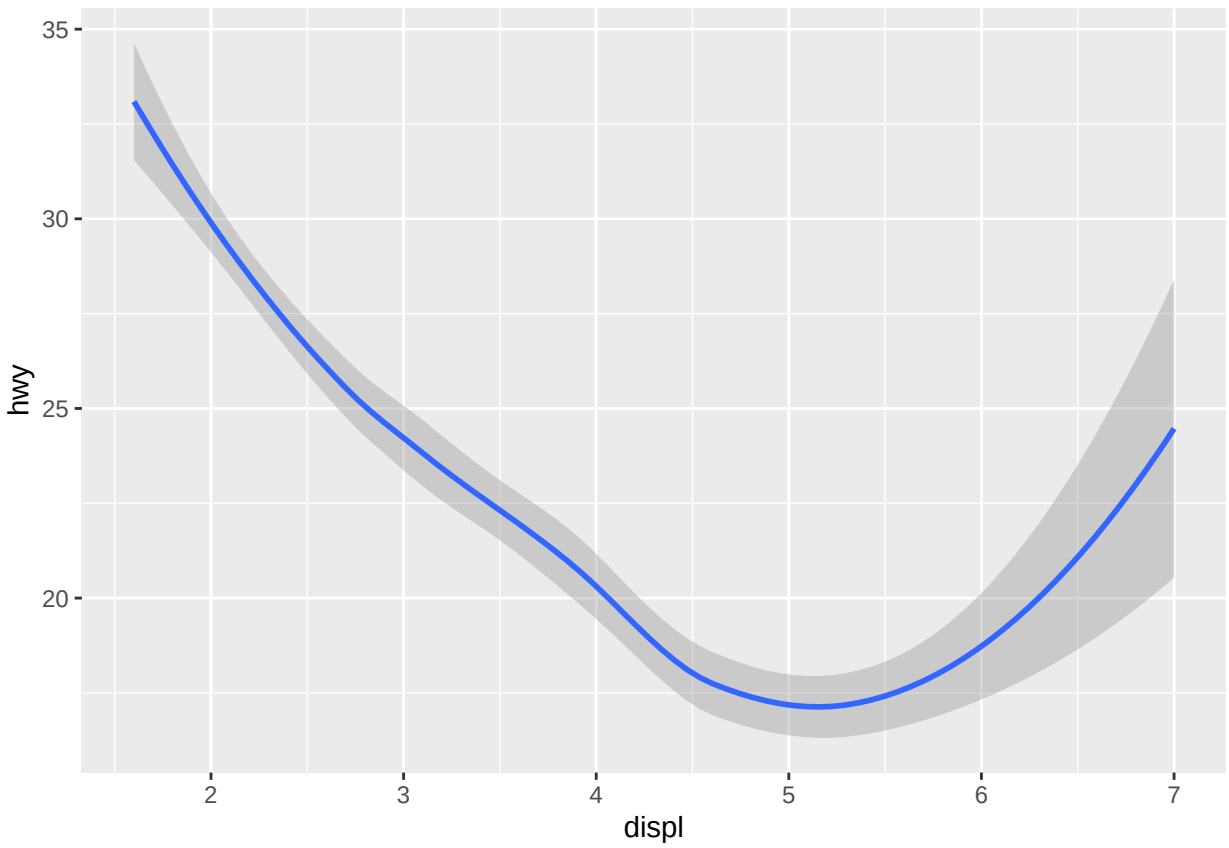



```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy,colour=drv))+scale_colour_viridis_d(option="plasma")+facet_gri
```



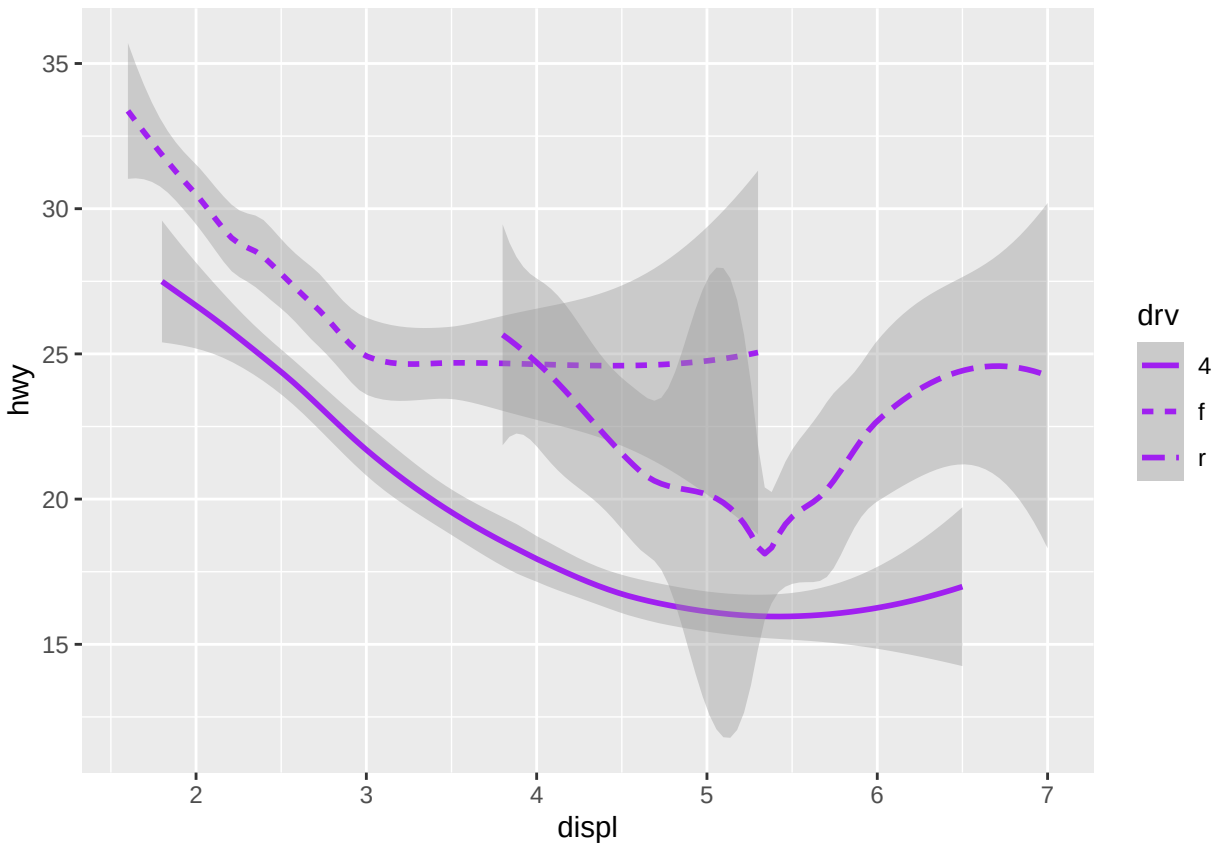

```
ggplot(mpg)+geom_smooth(aes(x=displ,y=hwy))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



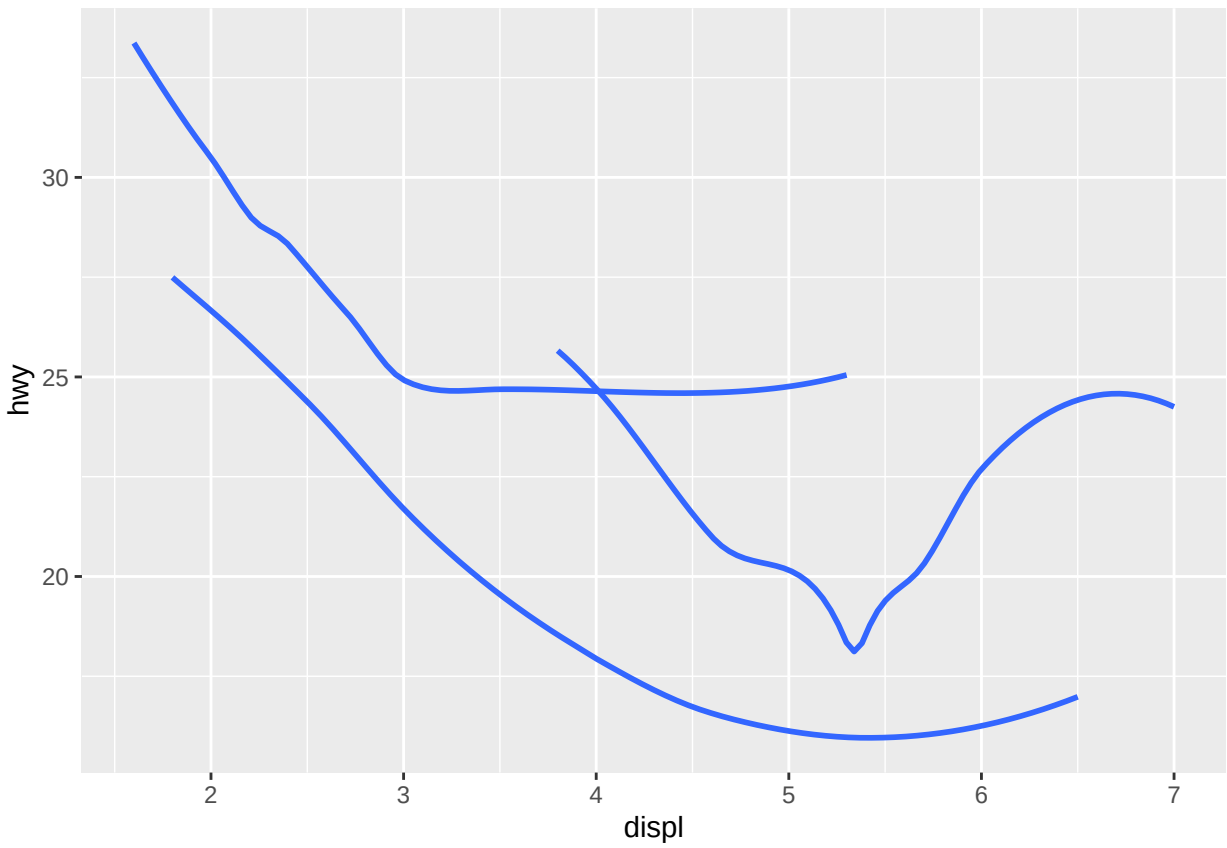

```r
ggplot(mpg)+geom_smooth(aes(x=displ,y=hwy,linetype=drv),colour="purple")
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



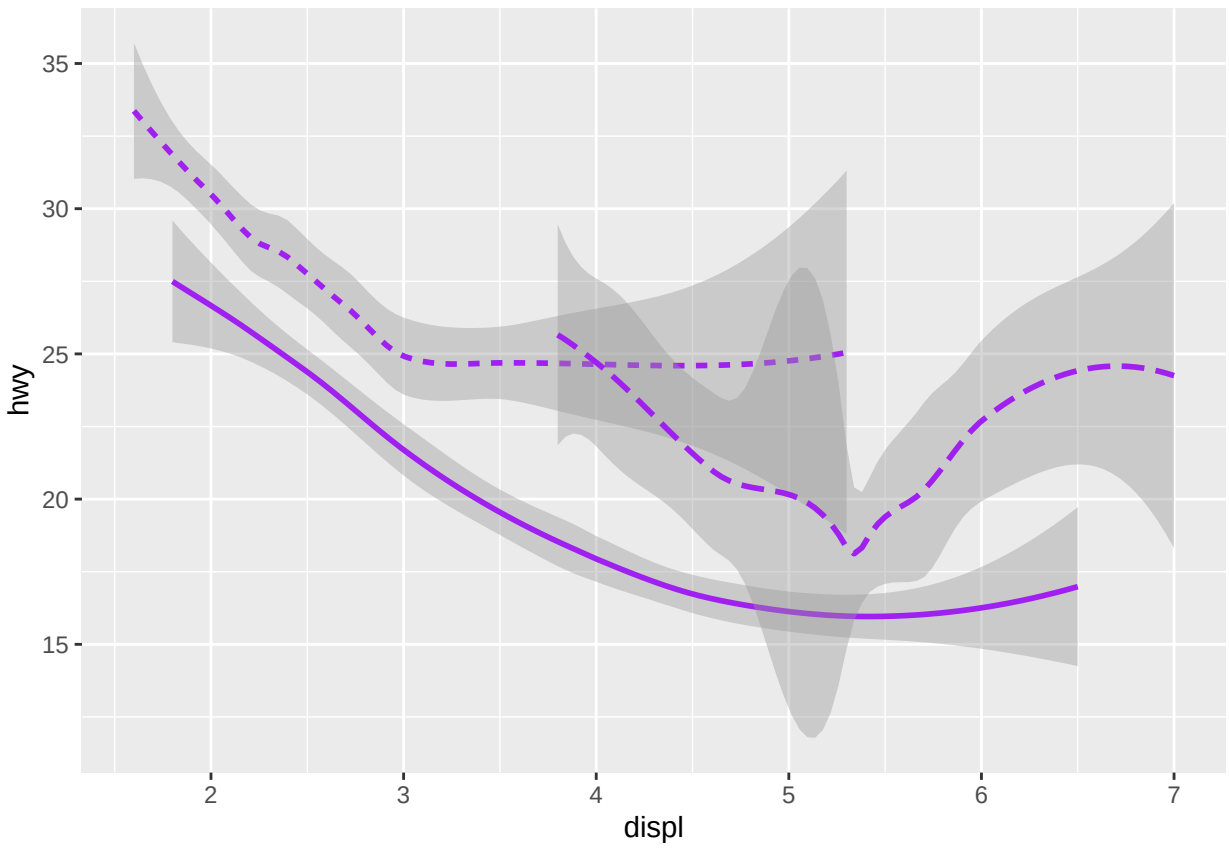

```r
ggplot(mpg)+geom_smooth(aes(x=displ,y=hwy,group=drv),se=FALSE)
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



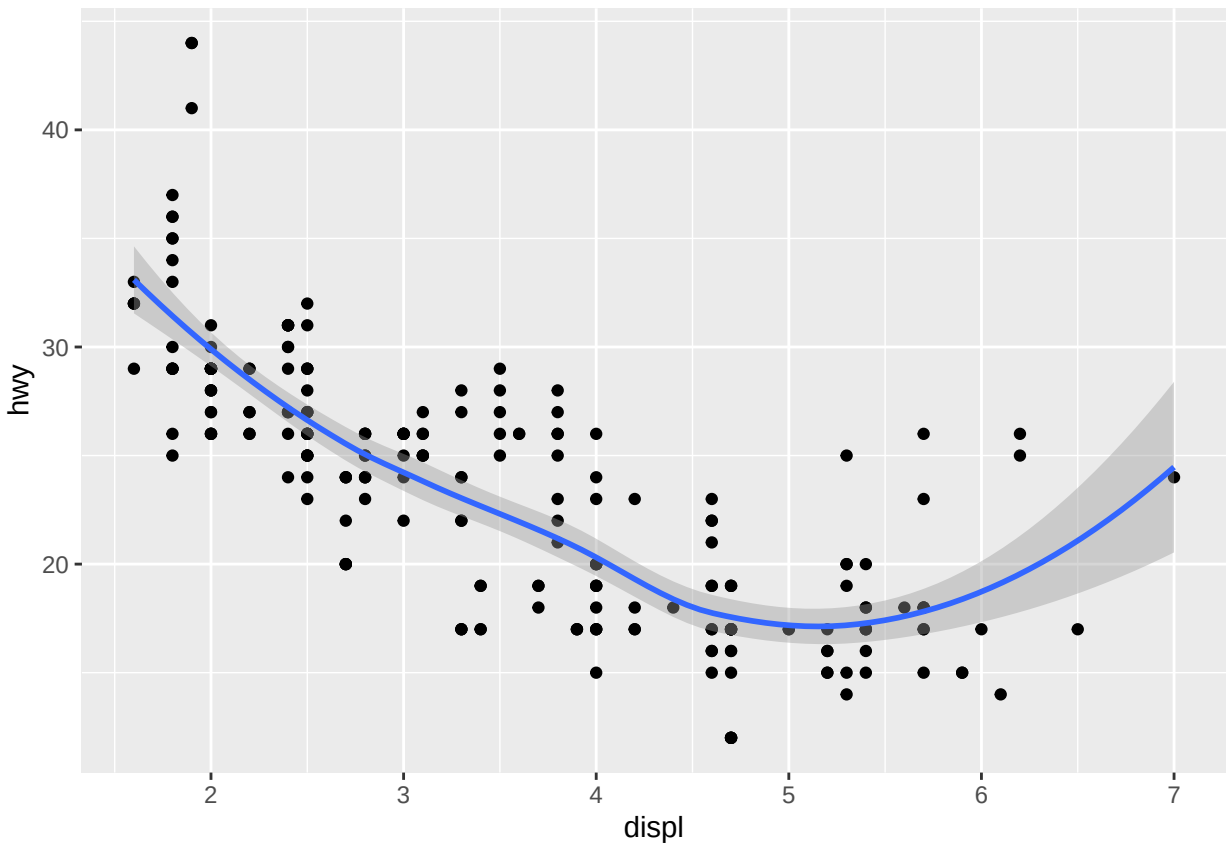

```r
ggplot(mpg)+geom_smooth(aes(x=displ,y=hwy,linetype=drv),colour="purple",show.legend=FALSE)
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

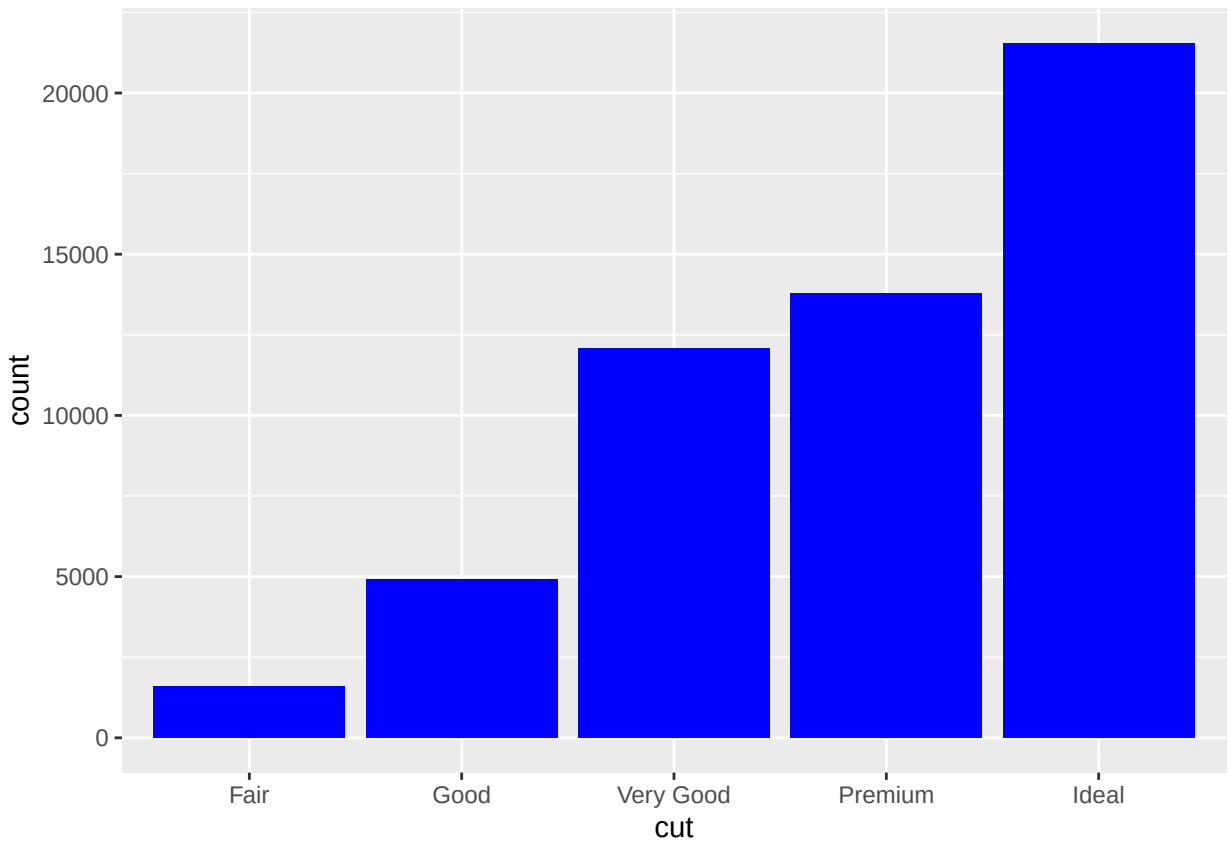



```r
ggplot(mpg)+geom_point(aes(x=displ,y=hwy))+geom_smooth(aes(x=displ,y=hwy))
```

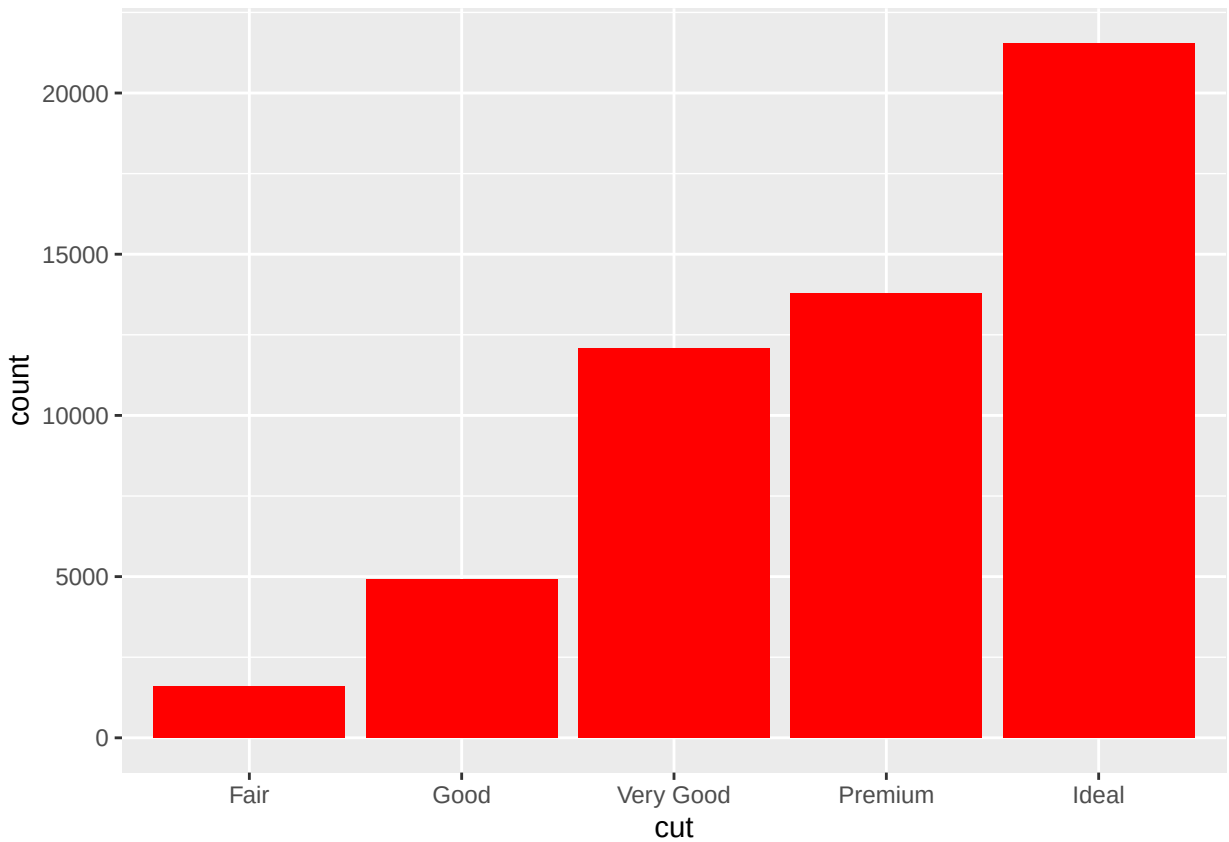
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



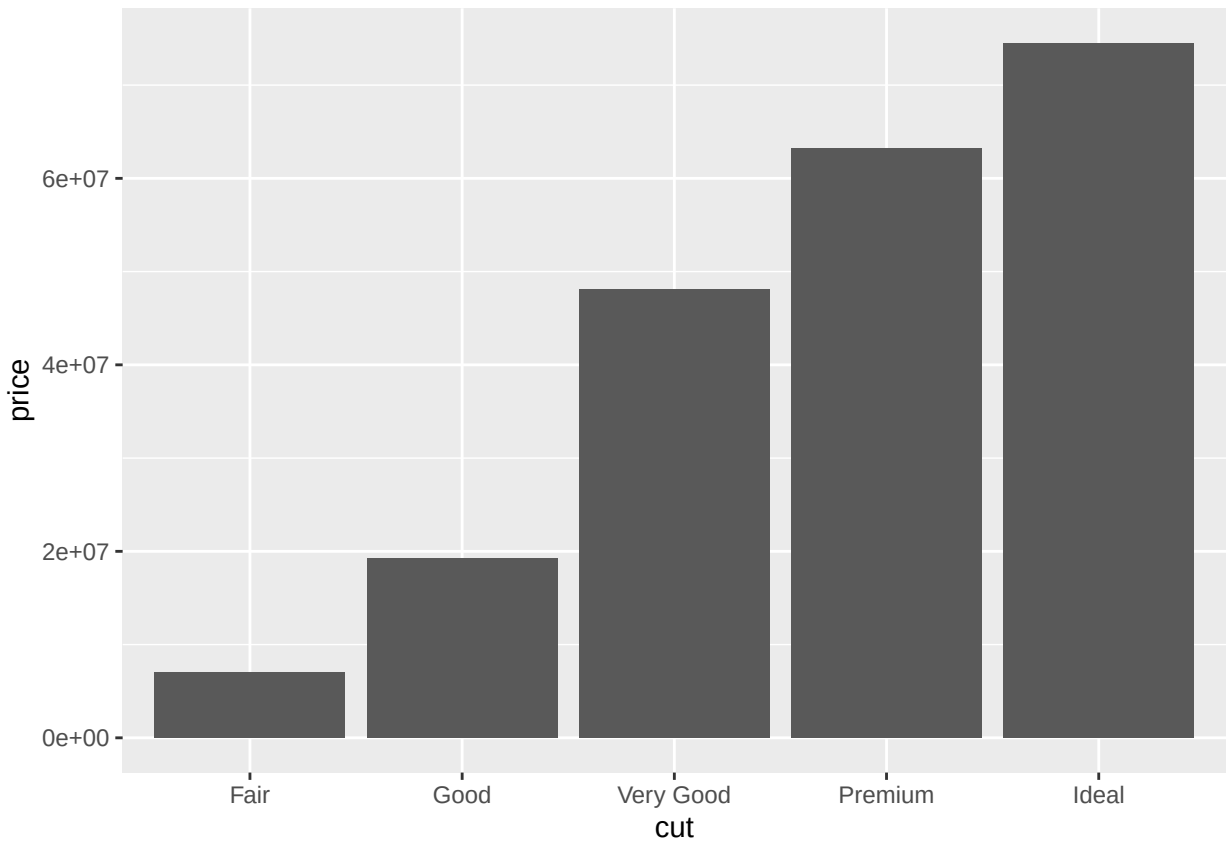

```
ggplot(diamonds)+geom_bar(aes(x=cut),fill="blue")
```



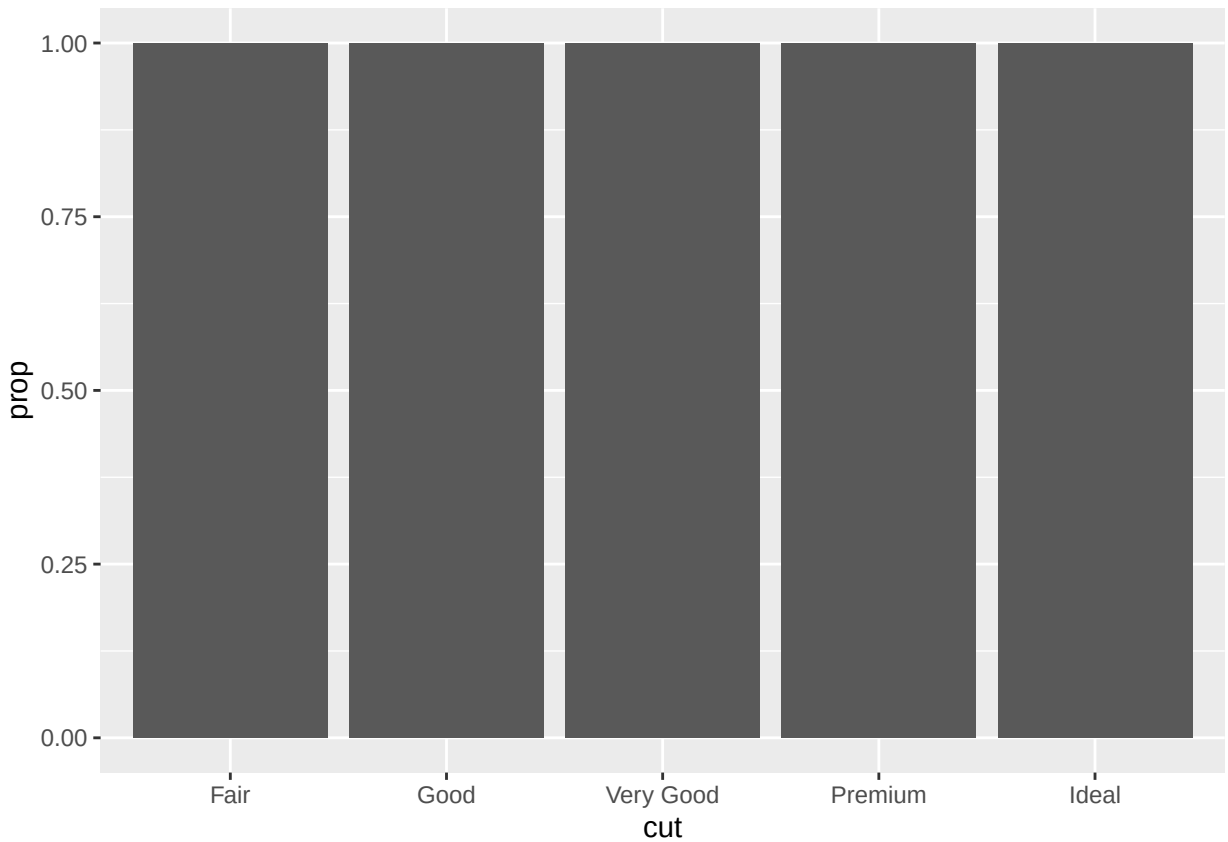

```
ggplot(diamonds)+stat_count(aes(x=cut),fill="red")
```



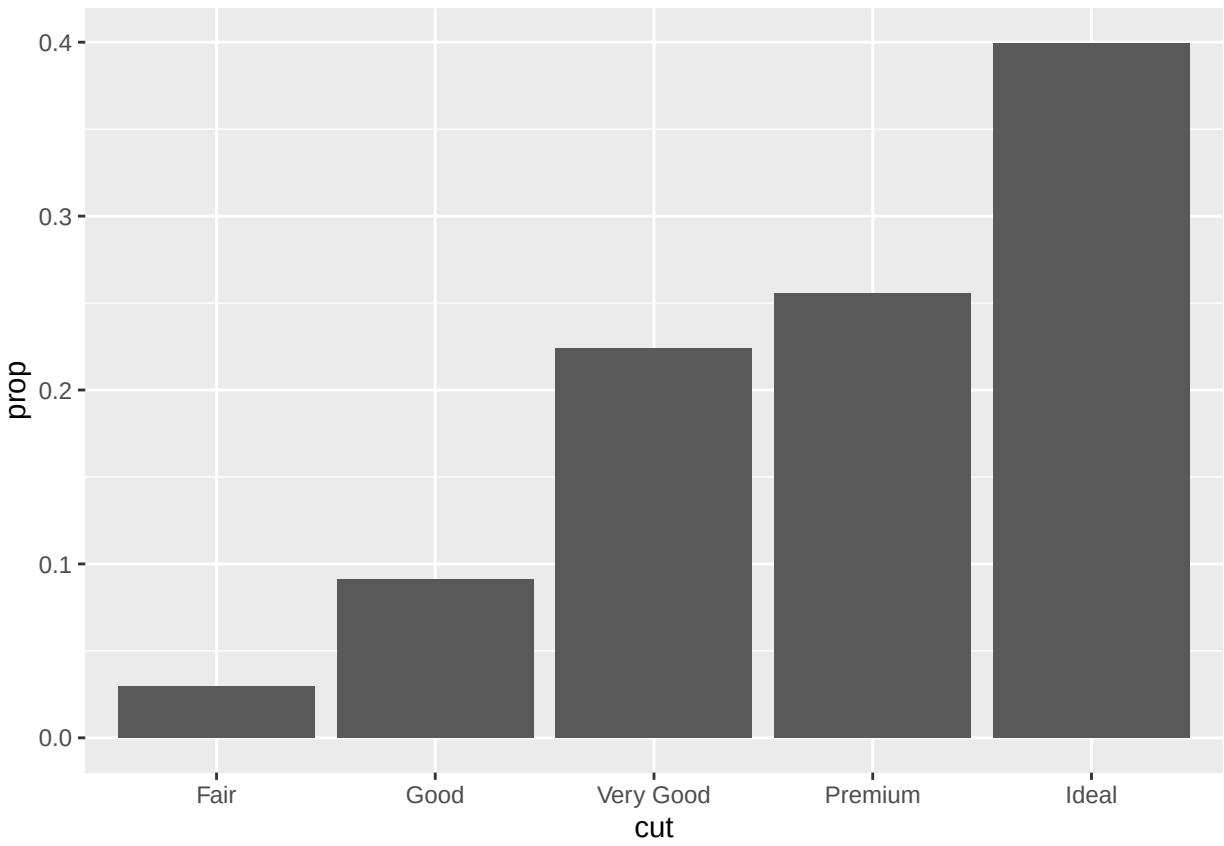

```
#Not mentioning a stat will cause error since stat_count only takes x or y values
ggplot(diamonds)+geom_bar(aes(x=cut,y=price),stat="identity")
```



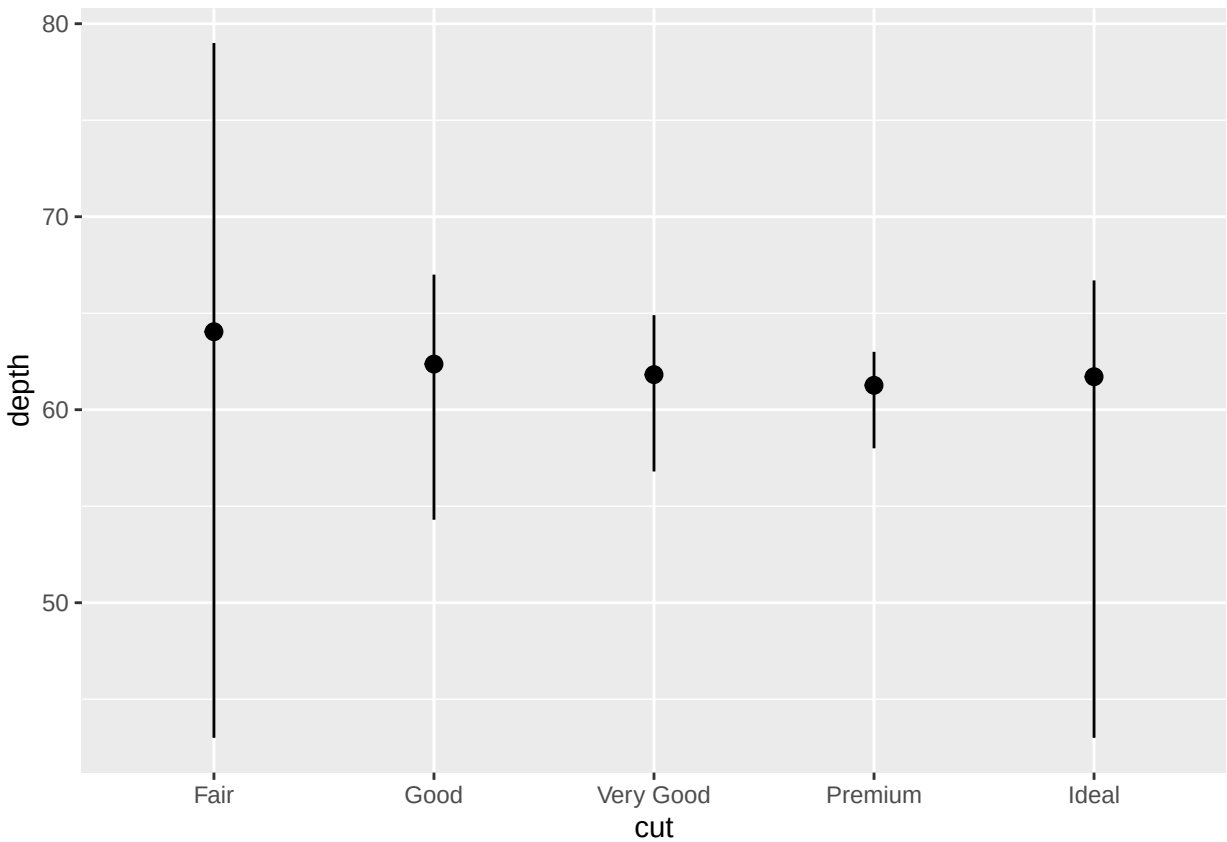

```
#This fails because ggplot considers each category as an individual group
#Naming a group fixes it as now ggplot considers the cut variable as one group
ggplot(diamonds)+geom_bar(aes(x=cut,y=after_stat(prop)))
```



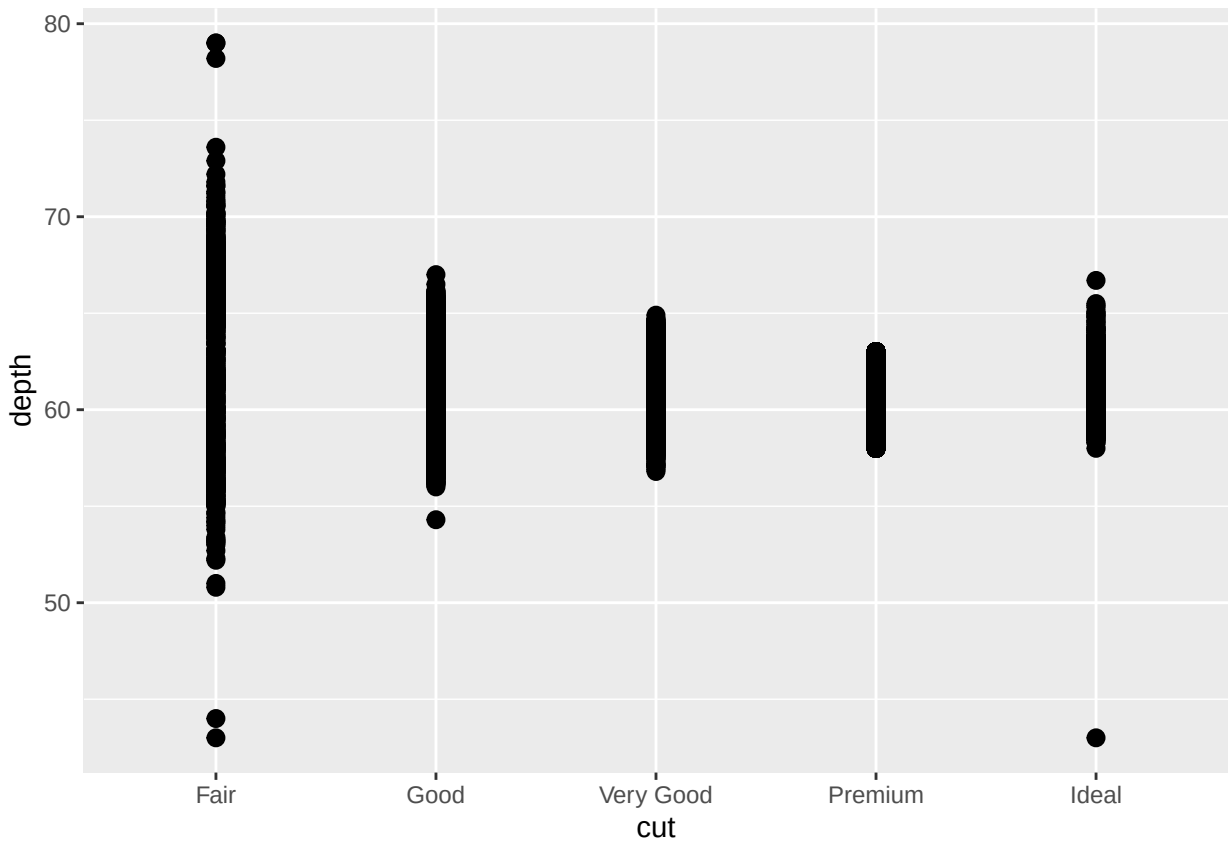

```
ggplot(diamonds)+geom_bar(aes(x=cut,y=after_stat(prop),group=1))
```



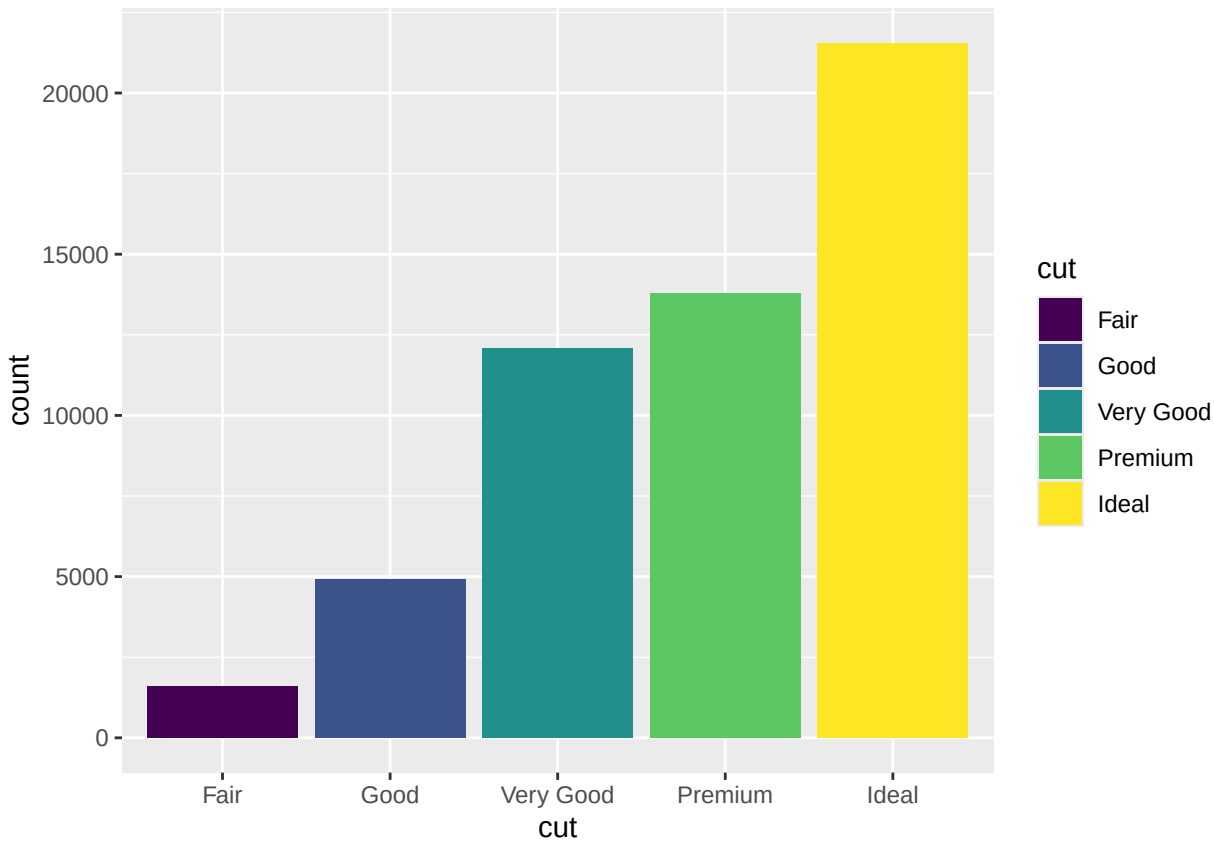

```
ggplot(diamonds)+stat_summary(aes(x=cut,y=depth),fun.min=min,fun.max=max,fun=mean)
```



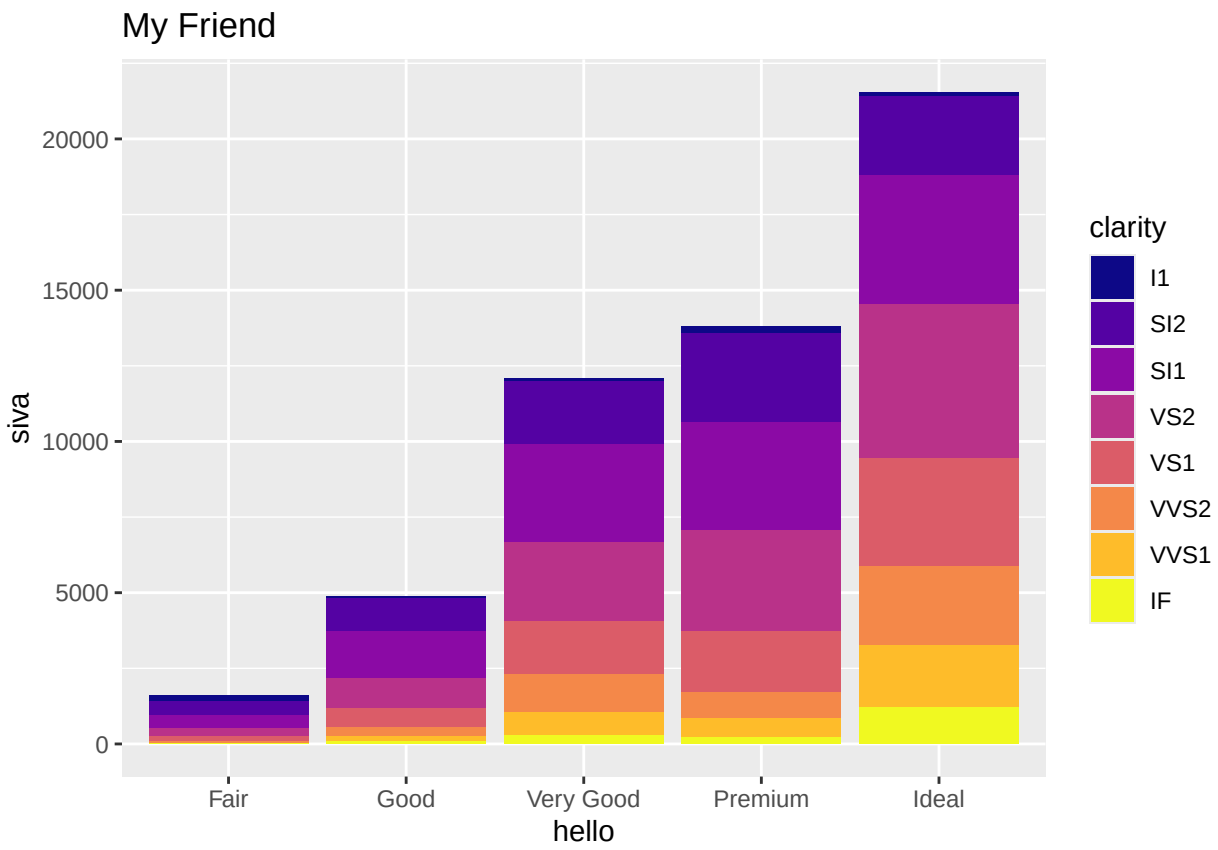

```
ggplot(diamonds)+geom_pointrange(aes(x=cut,y=depth,ymin=mean(depth)-sd(depth),ymax=mean(depth)+sd(depth
```



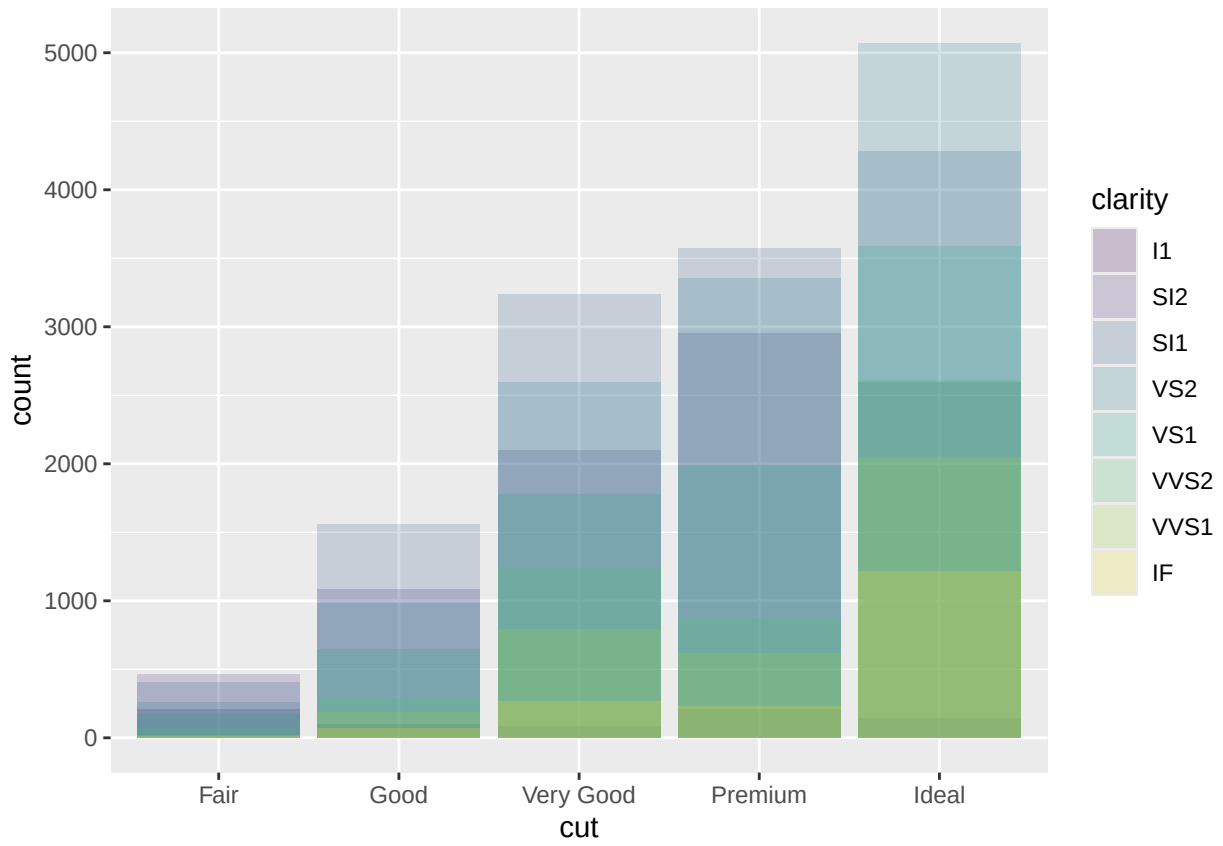

```
ggplot(diamonds)+geom_bar(aes(x=cut,fill=cut))
```



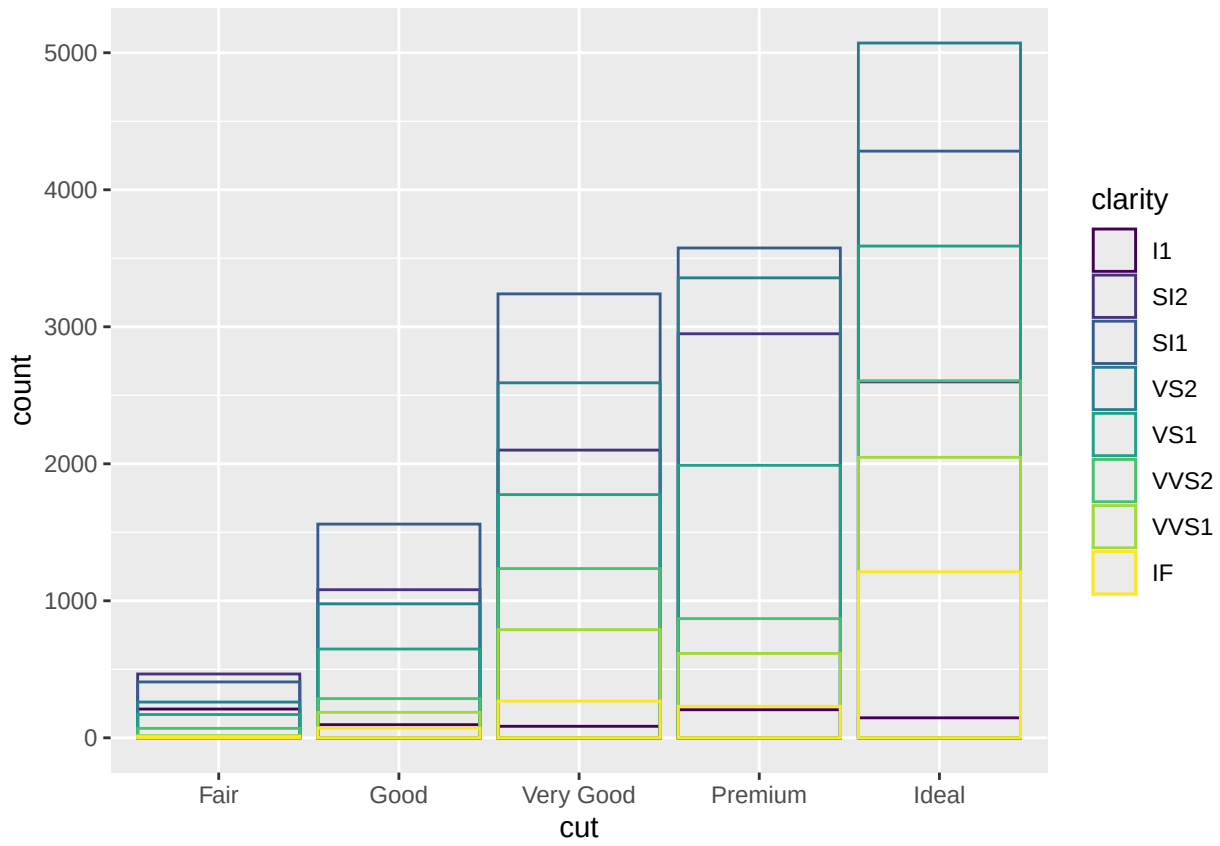

```
ggplot(diamonds)+geom_bar(aes(x=cut,fill=clarity))+scale_fill_viridis_d(option="plasma")+labs(x="hello"
```



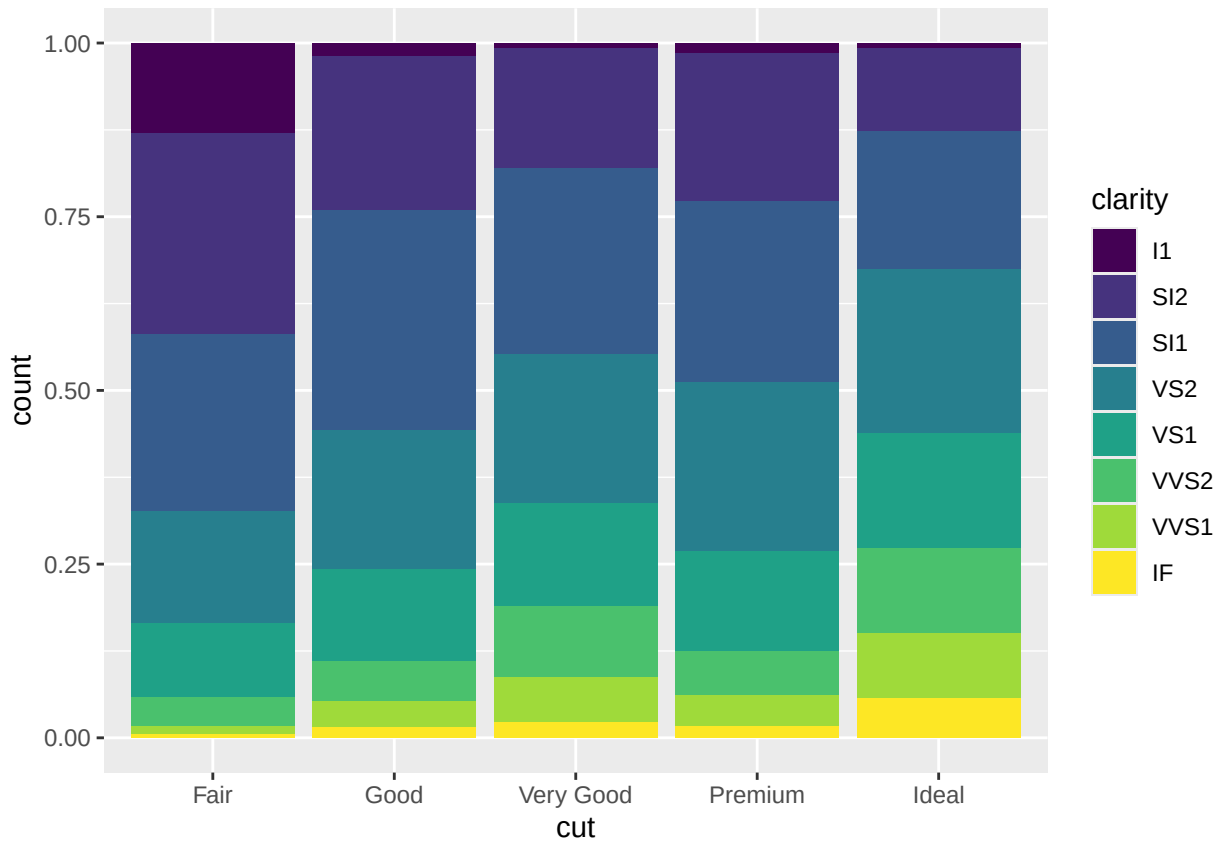

```
ggplot(diamonds,aes(x=cut,fill=clarity))+geom_bar(alpha=1/5,position="identity")
```



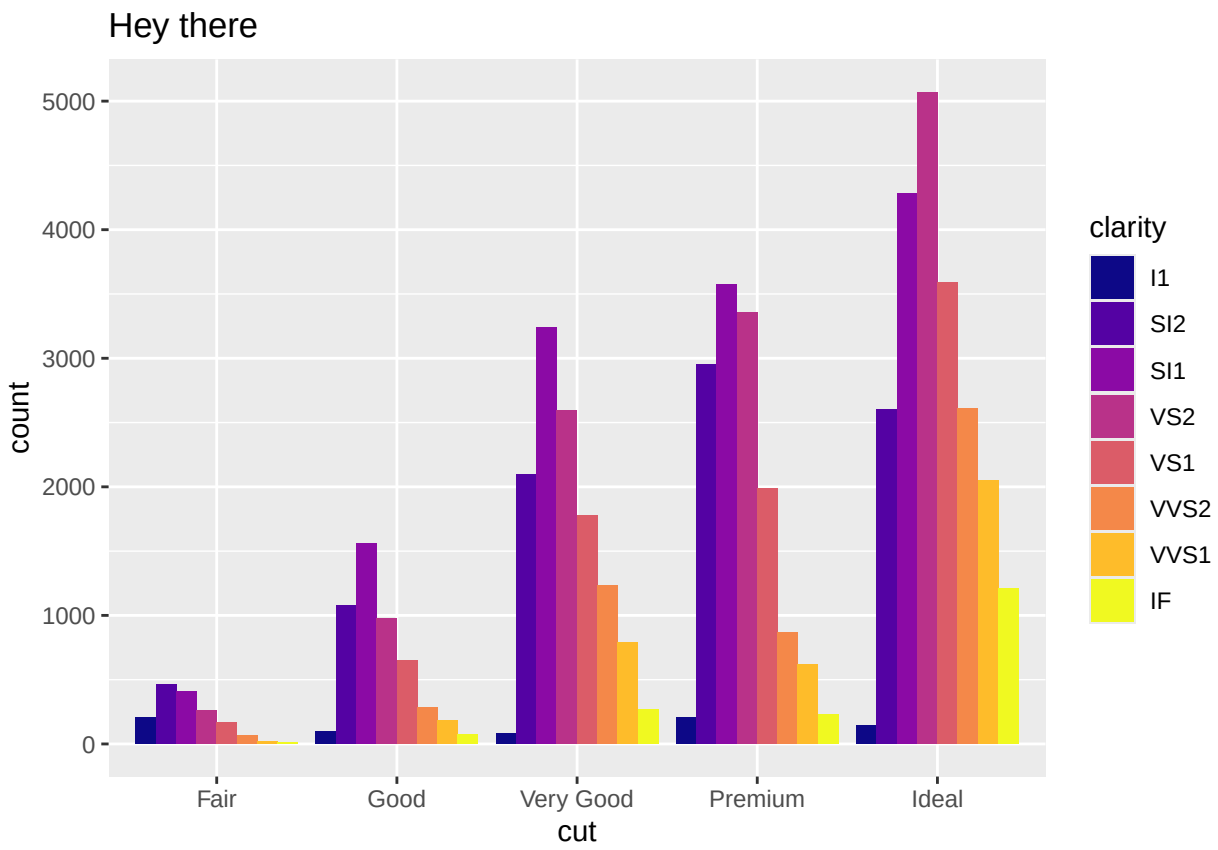

```
ggplot(diamonds,aes(x=cut,colour=clarity))+geom_bar(alpha=1/5,position="identity",fill=NA)
```



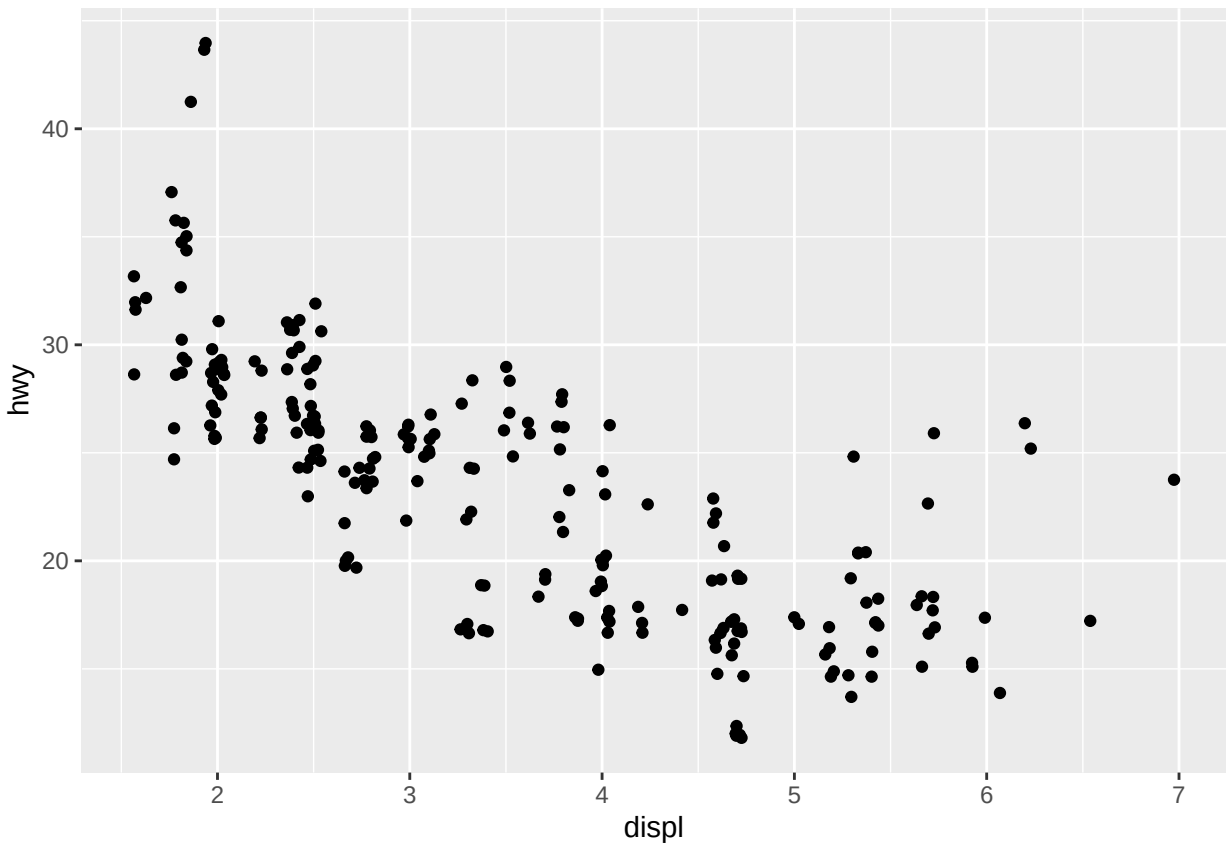

```
ggplot(diamonds)+geom_bar(aes(x=cut,fill=clarity),position="fill")
```



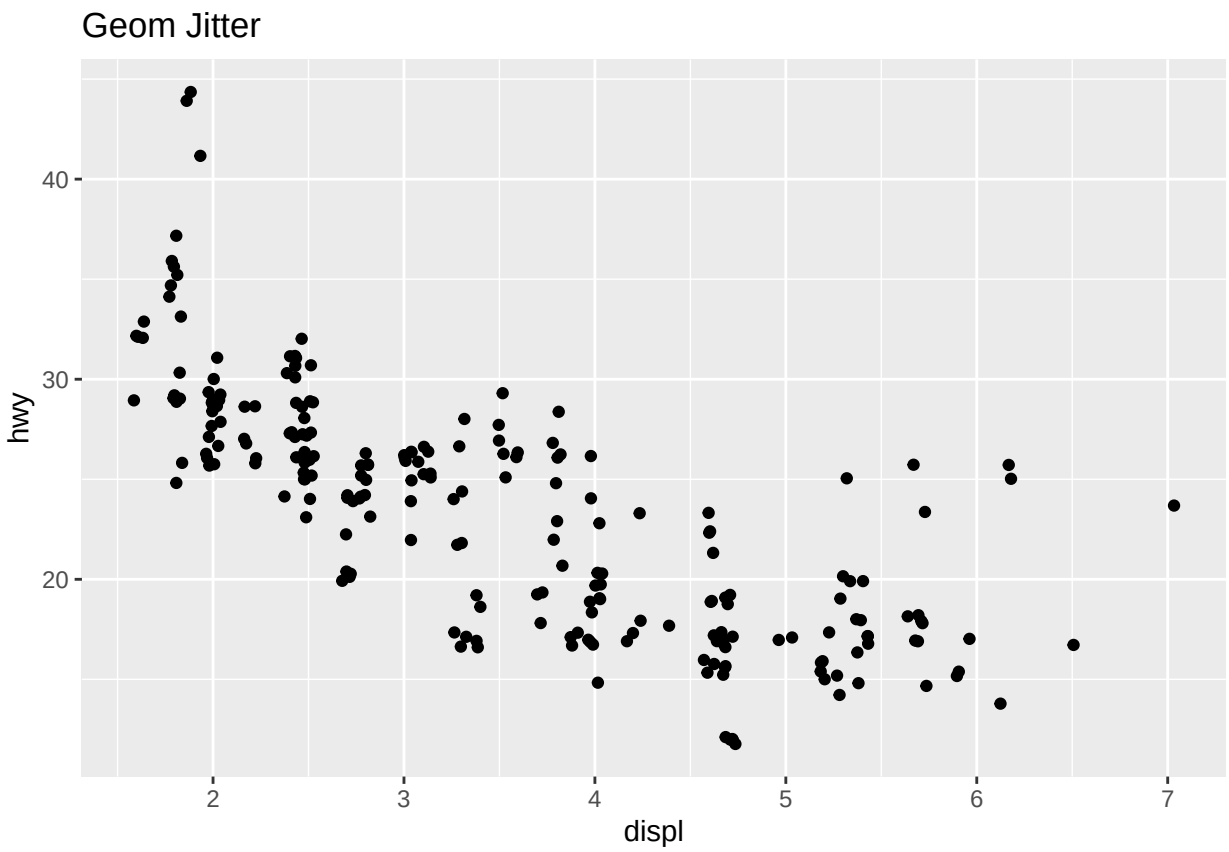

```
ggplot(diamonds)+geom_bar(aes(x=cut,fill=clarity),position="dodge")+scale_fill_viridis_d(option="plasma
```



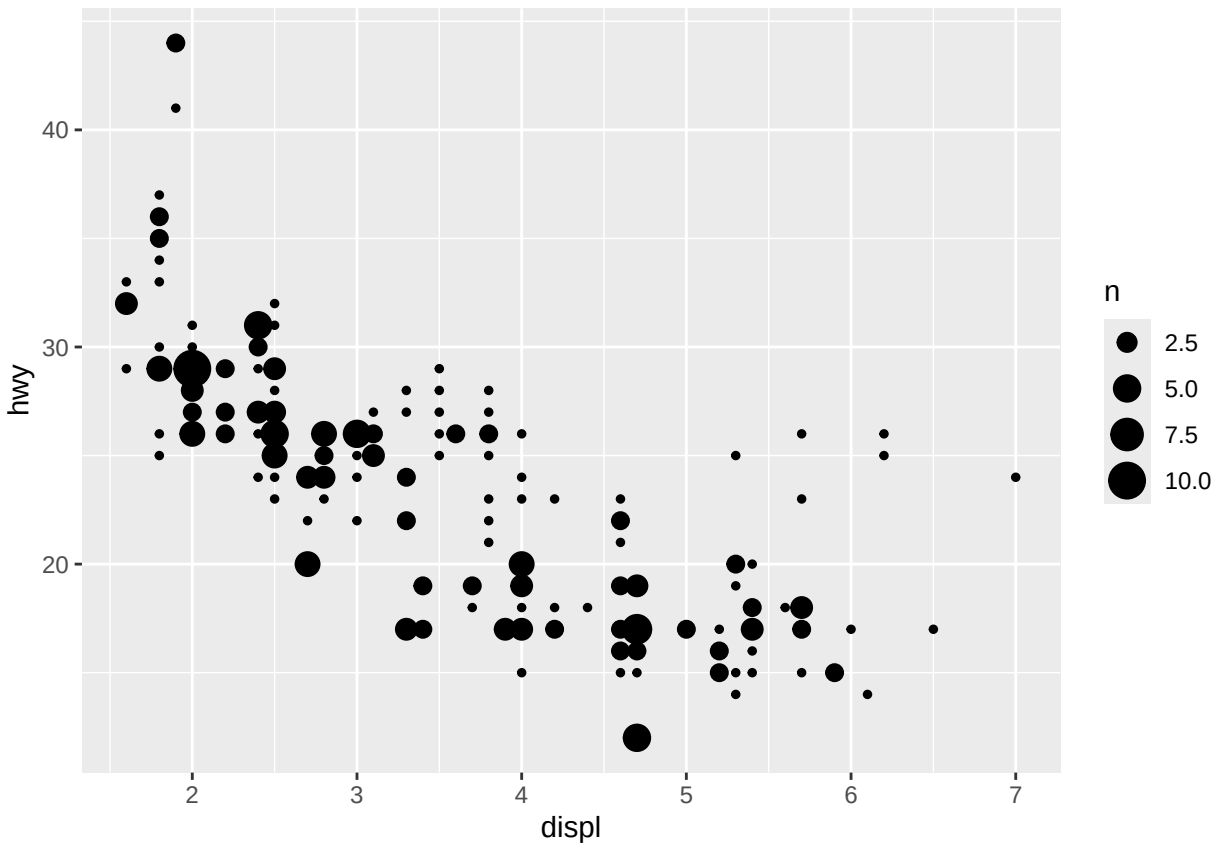

```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy),position="jitter")
```



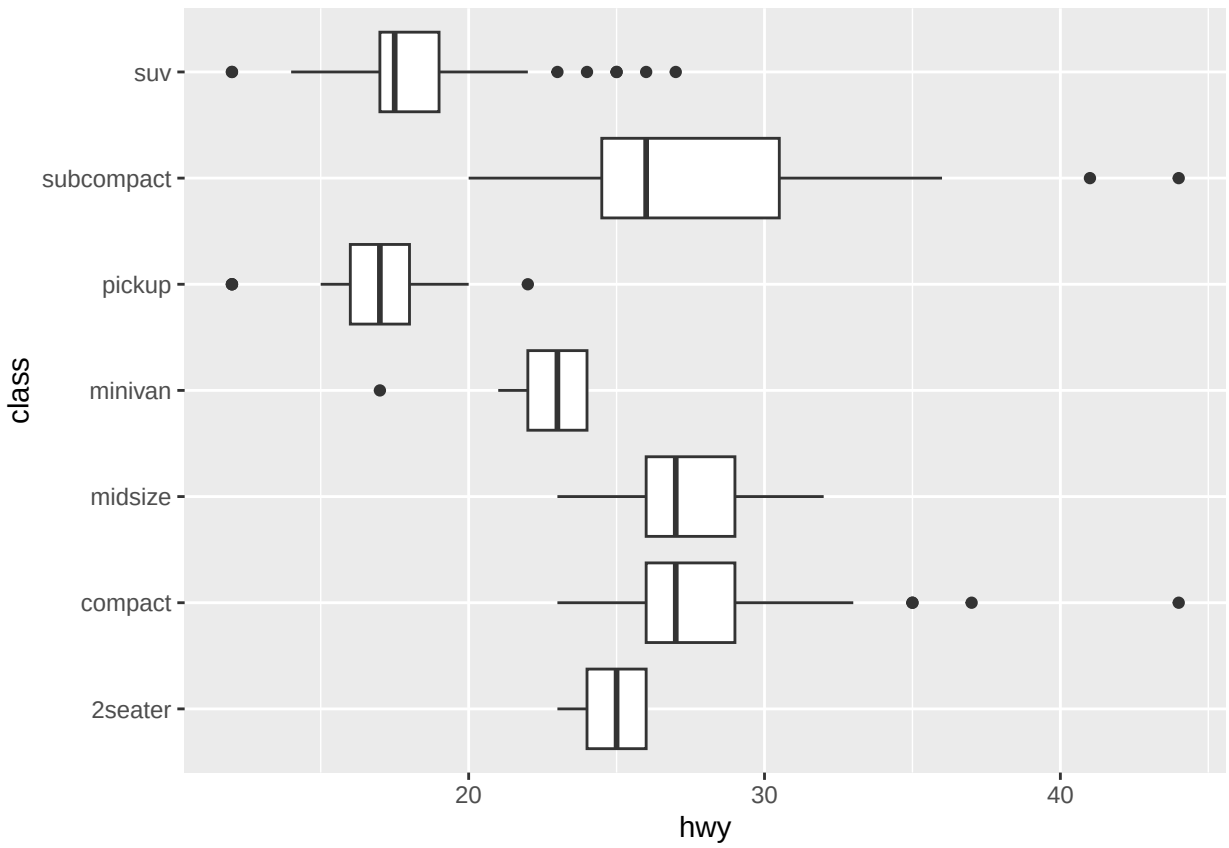

```
#One could also use geom_jitter that does exactly the same job
ggplot(mpg)+geom_jitter(aes(x=displ,y=hwy))+labs(title="Geom Jitter")
```




```
#geom_count denotes the size of a point by the number of overlaps, it uses stat sum
ggplot(mpg)+geom_count(aes(x=displ,y=hwy))
```

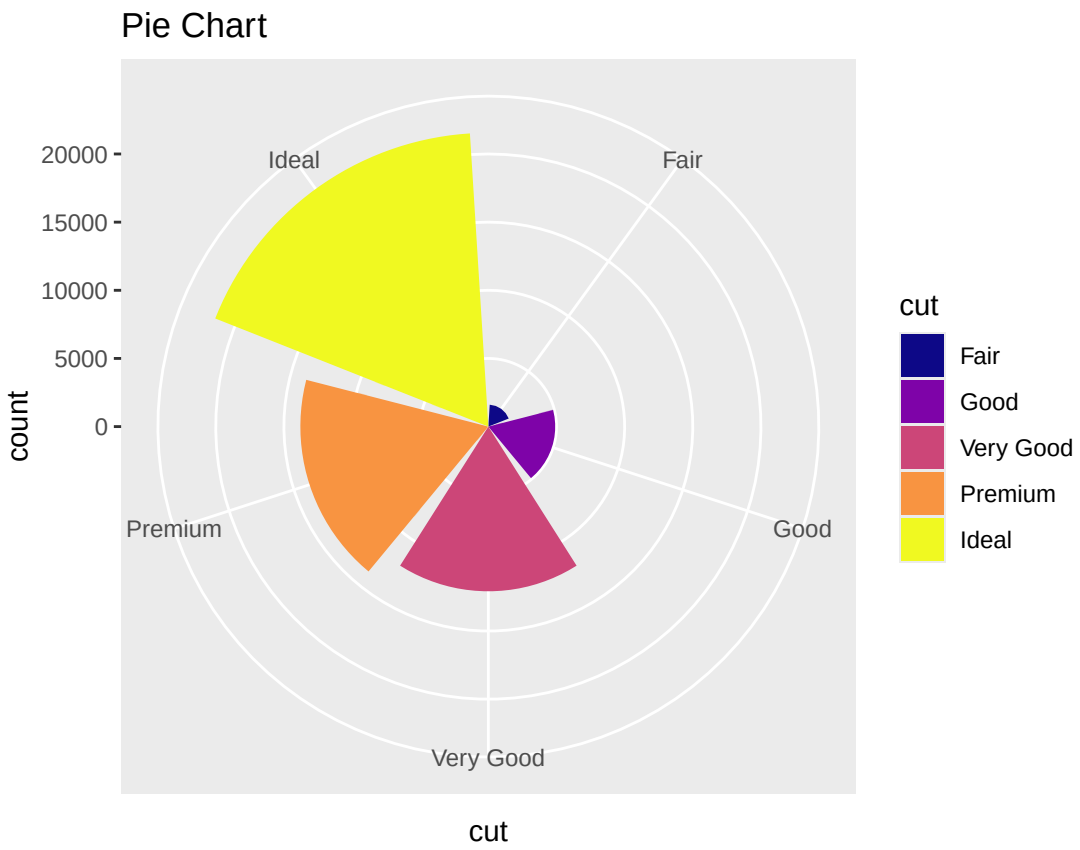



```
ggplot(mpg)+geom_boxplot(aes(x=class,y=hwy))+coord_flip()
```

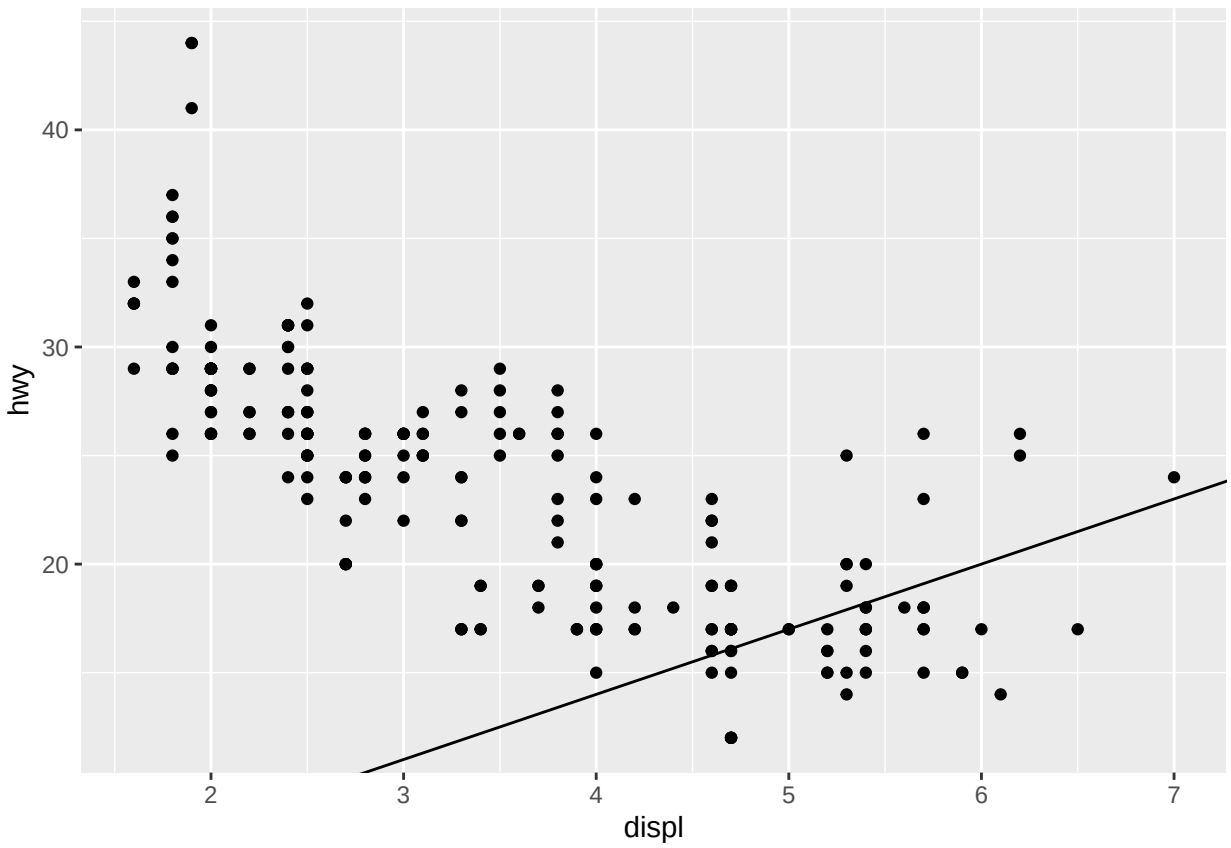



```
ggplot(diamonds)+geom_bar(aes(x=cut,fill=cut))+scale_fill_viridis_d(option="plasma")+coord_flip()+coord_
```

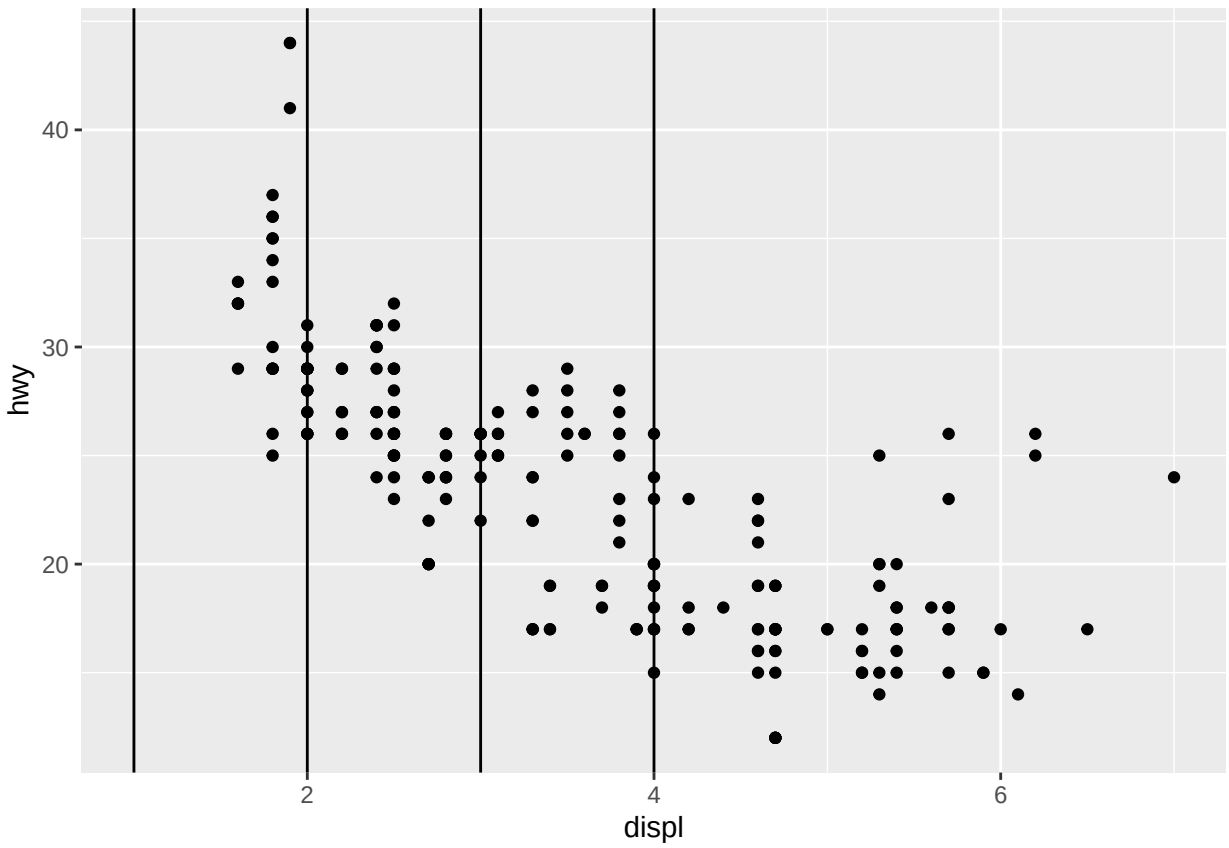
```
## Coordinate system already present.
## i Adding new coordinate system, which will replace the existing one.
```



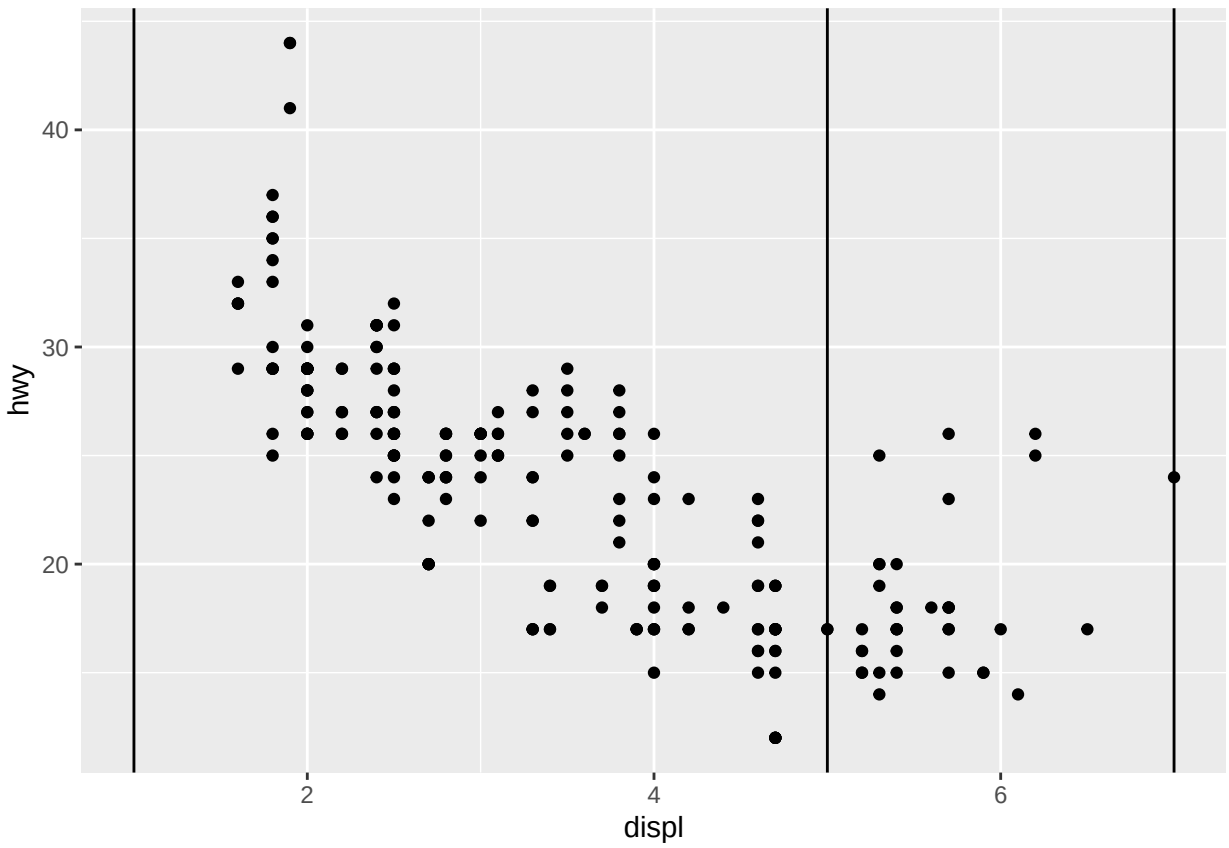

```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy))+geom_abline(slope=3,intercept=2)
```



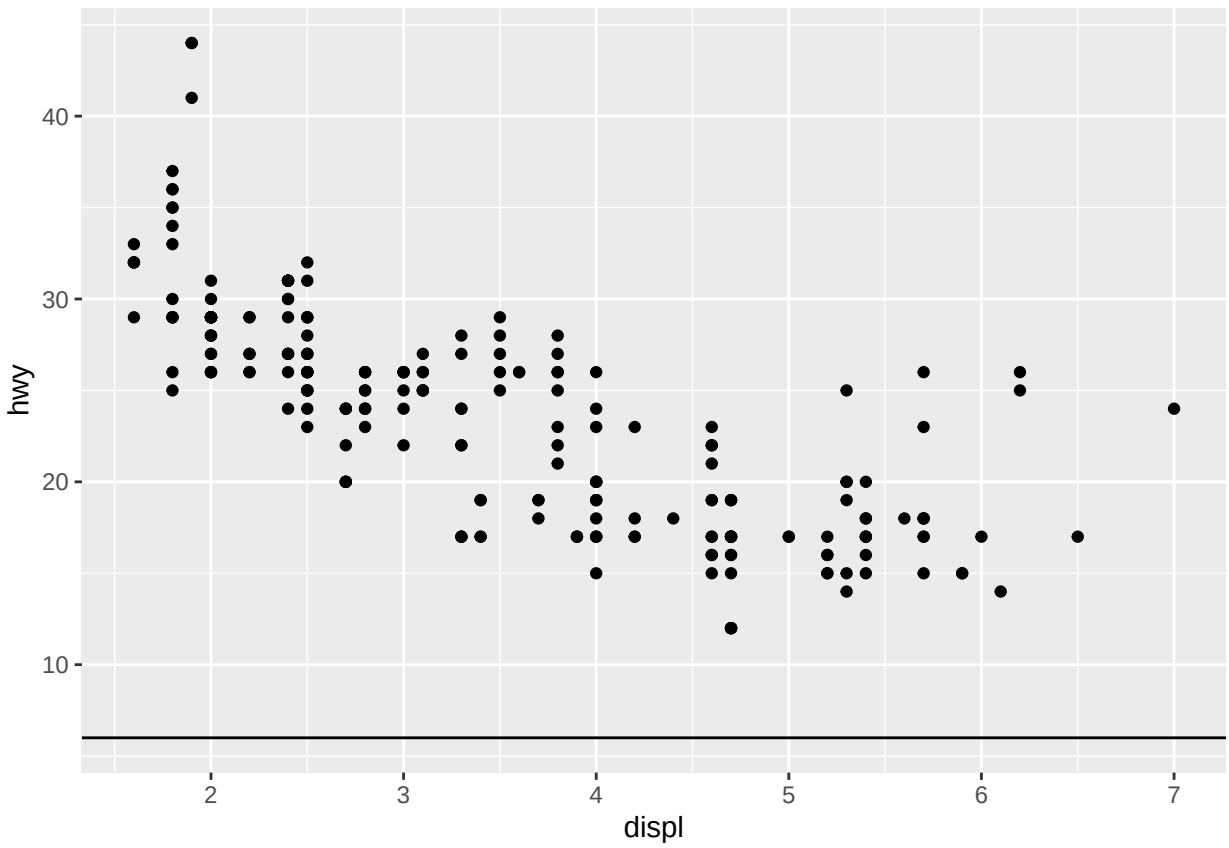

```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy))+geom_vline(xintercept=1:4)
```




```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy))+geom_vline(xintercept=c(1,5,7))
```




```
ggplot(mpg)+geom_point(aes(x=displ,y=hwy))+geom_hline(yintercept=6)
```




```r
#creating a dataframe
df=data.frame(
    name=c("A","B","C"),
    age=c(18,19,20),
    marks=c(80,75,90)
)
df
```

```
##   name age marks
## 1    A  18    80
## 2    B  19    75
## 3    C  20    90
```

```r
str(df)
```

```
## 'data.frame':    3 obs. of  3 variables:
##  $ name : chr  "A" "B" "C"
##  $ age  : num  18 19 20
##  $ marks: num  80 75 90
```

```r
x=1:3
y=c("low","medium","high")
df1=data.frame(x,y)
df1
```

```
##   x      y
## 1 1    low
## 2 2 medium
## 3 3   high
```

```r
#From a matrix
mat=matrix(1:6,nrow=3)
df2=as.data.frame(mat)
df2
```

```
##   V1 V2
## 1  1  4
## 2  2  5
## 3  3  6
```

```r
df$name
```

```
## [1] "A" "B" "C"
```

```r
#Accessing by index
df[2,]
```

```
##   name age marks
## 2    B  19    75
```

```r
df[2,3]
```

```
## [1] 75
```

```r
df$gender=c("M","F","friend")
df
```

```
##   name age marks gender
## 1    A  18    80      M
## 2    B  19    75      F
## 3    C  20    90 friend
```

```r
df$name[2]="Hello"
df
```

```
##    name age marks gender
## 1     A  18    80      M
## 2 Hello  19    75      F
## 3     C  20    90 friend
```

```r
df[df$age>18]
```

```
##   age marks
## 1  18    80
## 2  19    75
## 3  20    90
```

```r
df[df$name=="A"]
```

```
##    name gender
## 1     A      M
## 2 Hello      F
## 3     C friend
```

```r
df$gender=NULL
df1=data.frame(
    name=c("A","B"),
    marks=c(80,75)
)
df2=data.frame(
```

```r
    name=c("C","D"),
    marks=c(90,85)
)
rbind(df1,df2)
```

```
##   name marks
## 1    A    80
## 2    B    75
## 3    C    90
## 4    D    85
```

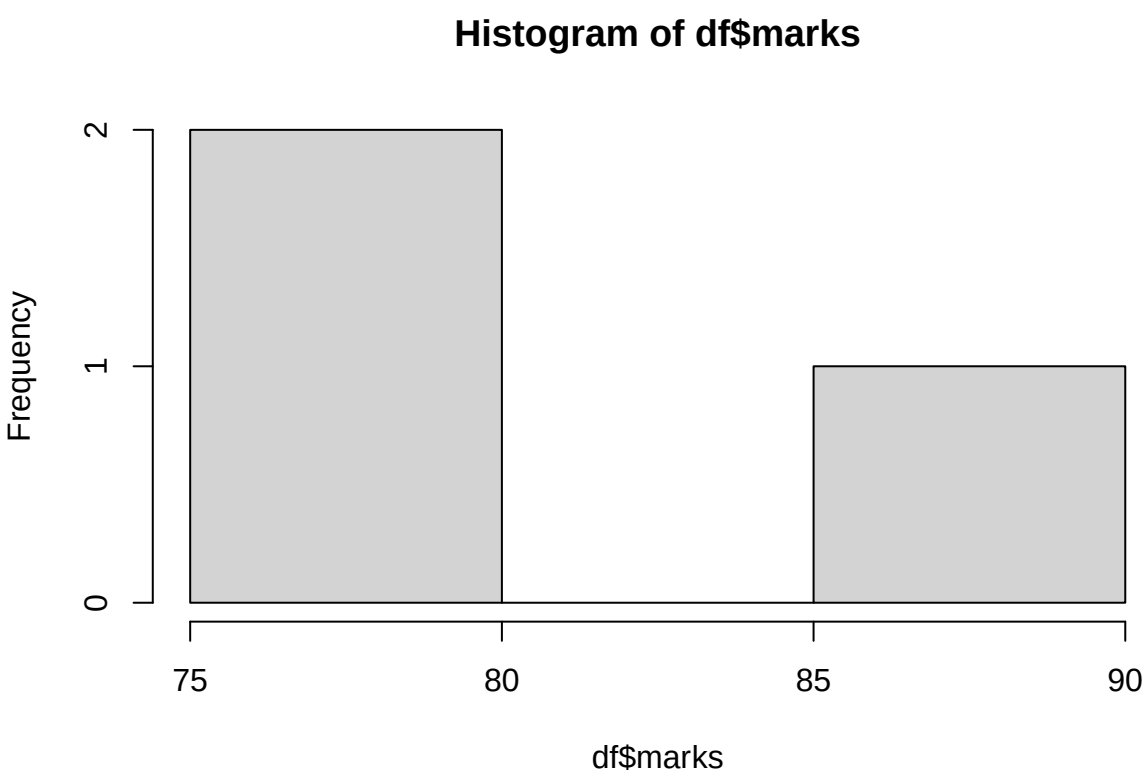
```r
df1
```

```
##   name marks
## 1    A    80
## 2    B    75
```

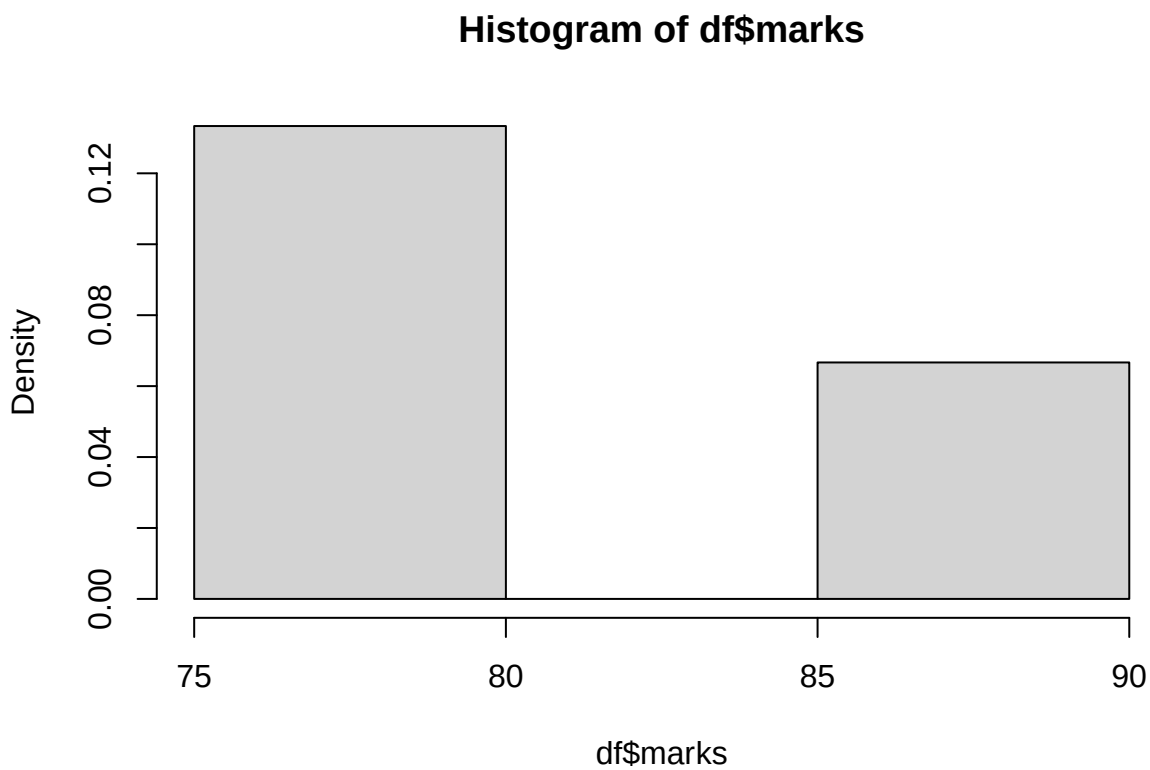
```r
df2
```

```
##   name marks
## 1    C    90
## 2    D    85
```

```r
#Factors are used to create categorical variables and levels are the numeric labels under them
df$group=factor(c("Control","Treatment","Control"),levels=c("Control","Treatment"))
hist(df$marks)
```



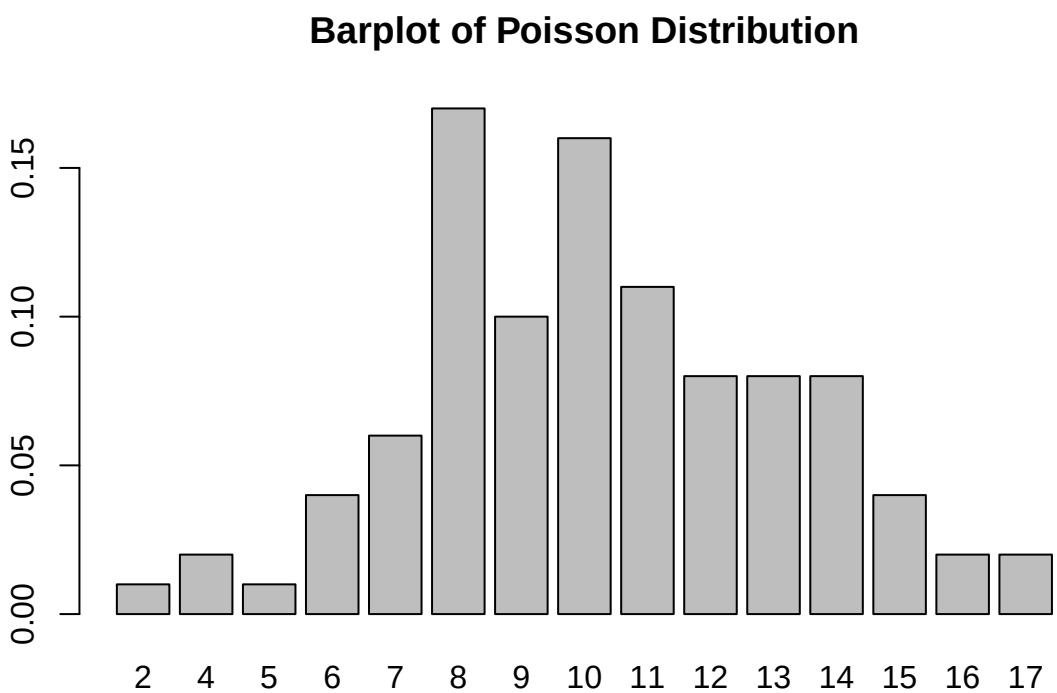

```r
hist(df$marks,probability=TRUE)
```



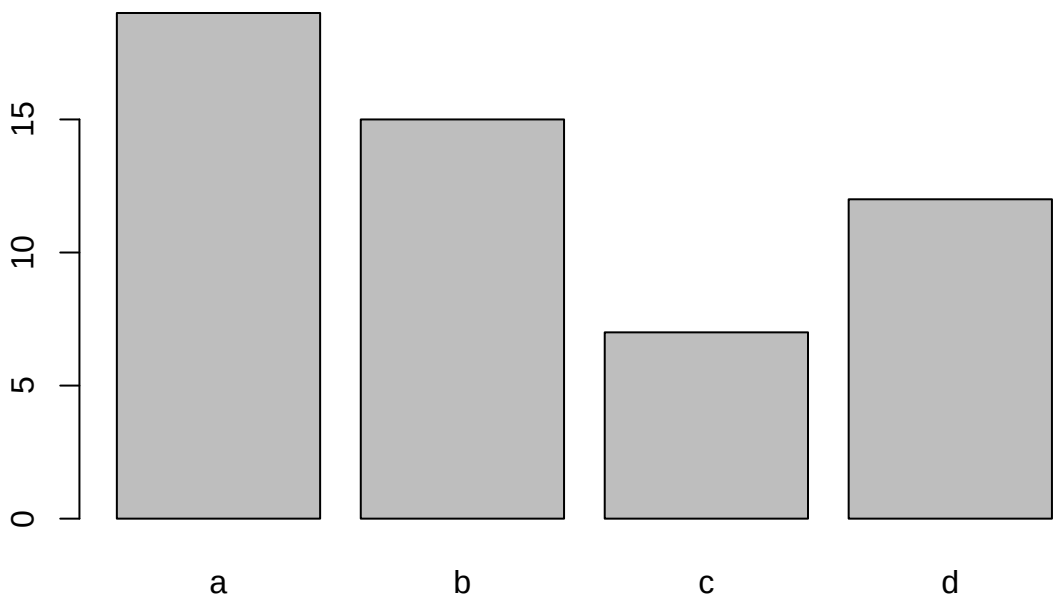

```
df
```

```
##    name age marks     group
## 1     A  18    80   Control
## 2 Hello  19    75 Treatment
## 3     C  20    90   Control
```

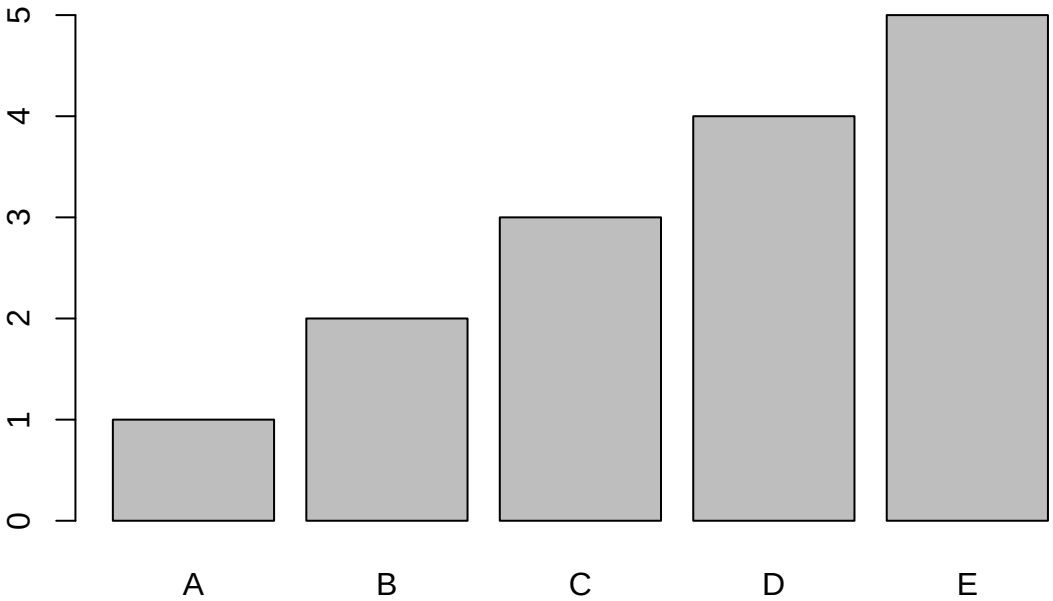
```
x=rpois(100,10)
#Barplot only takes in height values
barplot(height=prop.table(table(x)),main="Barplot of Poisson Distribution")
```




```
count=c(a=19,b=15,c=7,d=12)
barplot(count)
```




```
#We could also add names like this:
barplot(c(1,2,3,4,5),names.arg=c("A","B","C","D","E"))
```




```r
prop.table(table(x))
```

```
## x
##    2    4    5    6    7    8    9   10   11   12   13   14   15   16   17
## 0.01 0.02 0.01 0.04 0.06 0.17 0.10 0.16 0.11 0.08 0.08 0.08 0.04 0.02 0.02
```

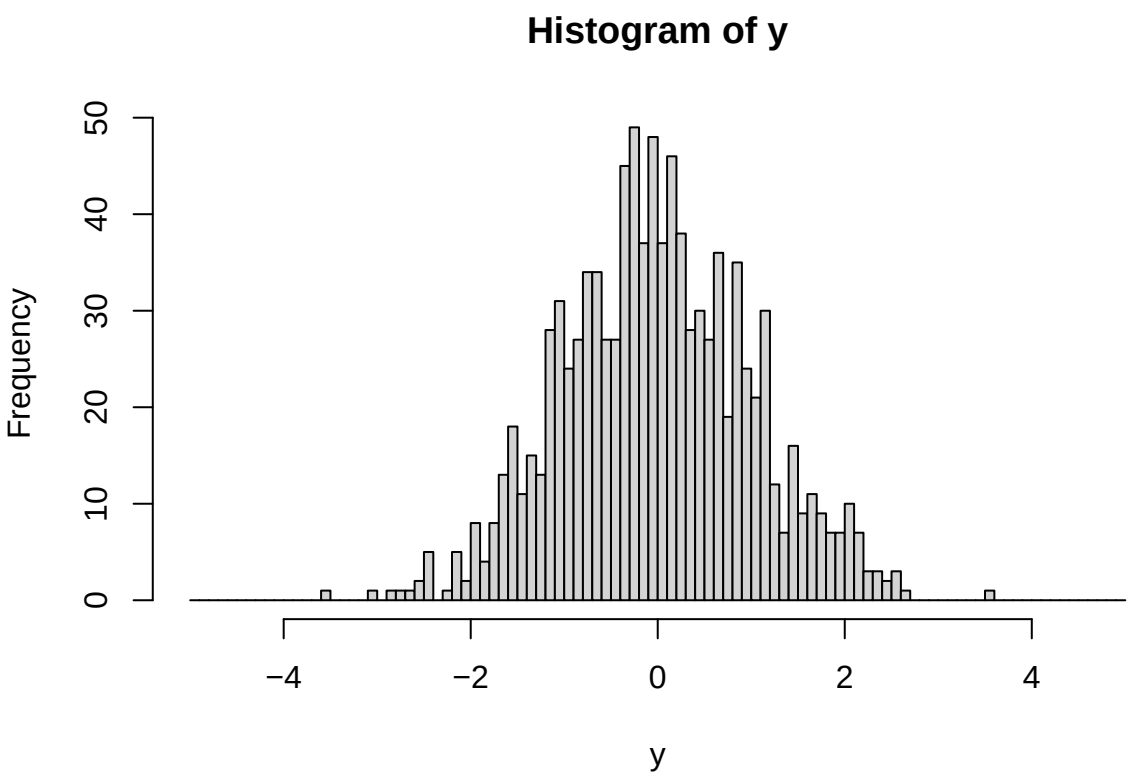
```r
#Creating a matrix in R
m=matrix(1:9,nrow=3,ncol=3)
m
```

```
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
```

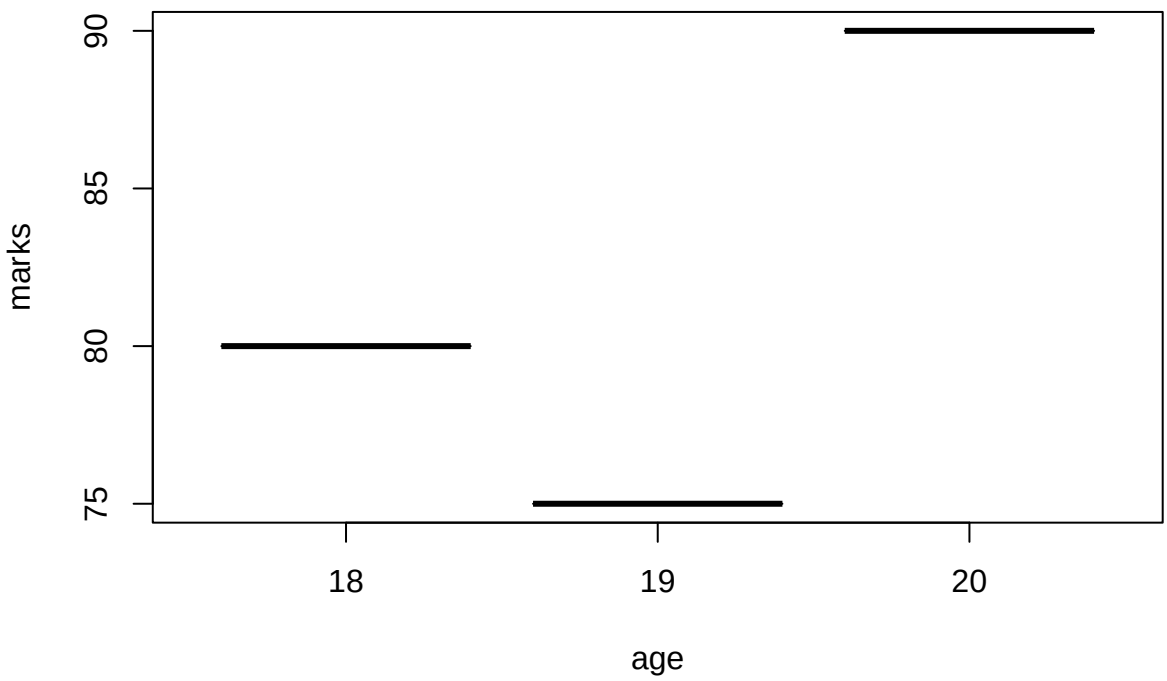
```r
#By default matrices in R are column major, we could however also fill them by rows
m=matrix(1:9,nrow=3,ncol=3,byrow=TRUE)
m
```

```
##      [,1] [,2] [,3]
## [1,]    1    2    3
## [2,]    4    5    6
## [3,]    7    8    9
```

```r
y=rnorm(1000)
hist(y,breaks=seq(-5,5,by=0.1))
```

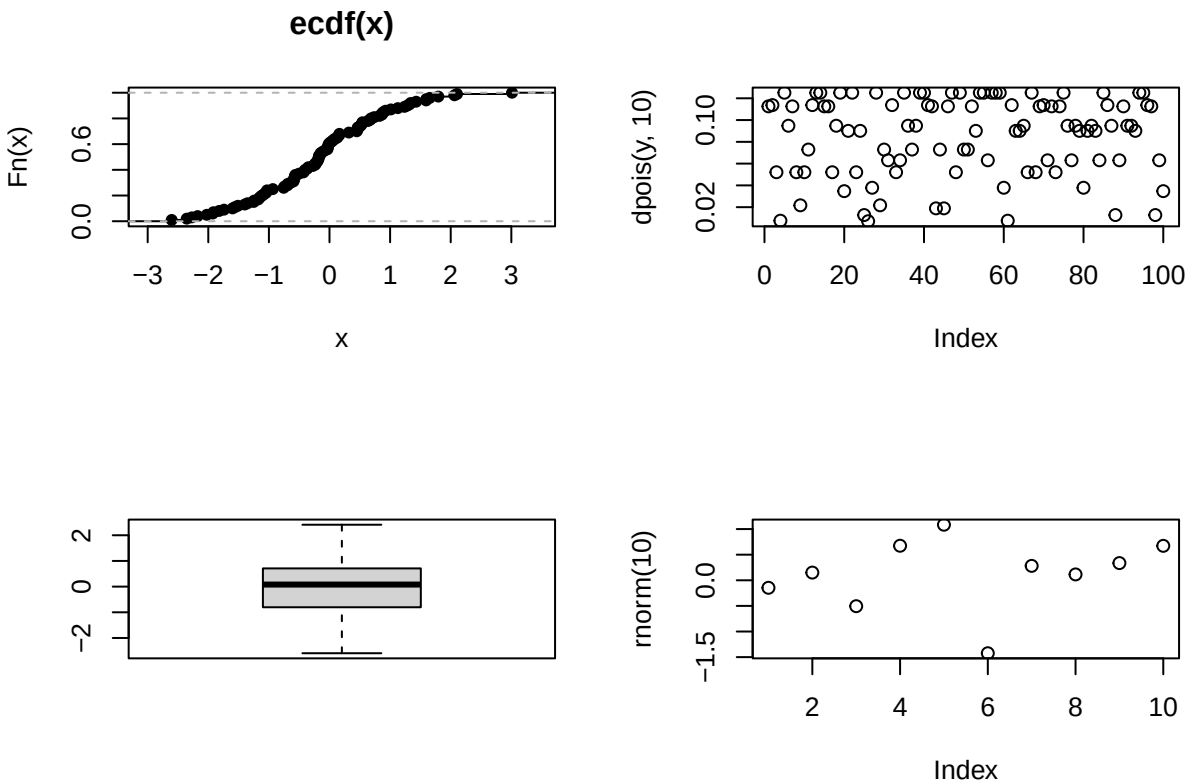


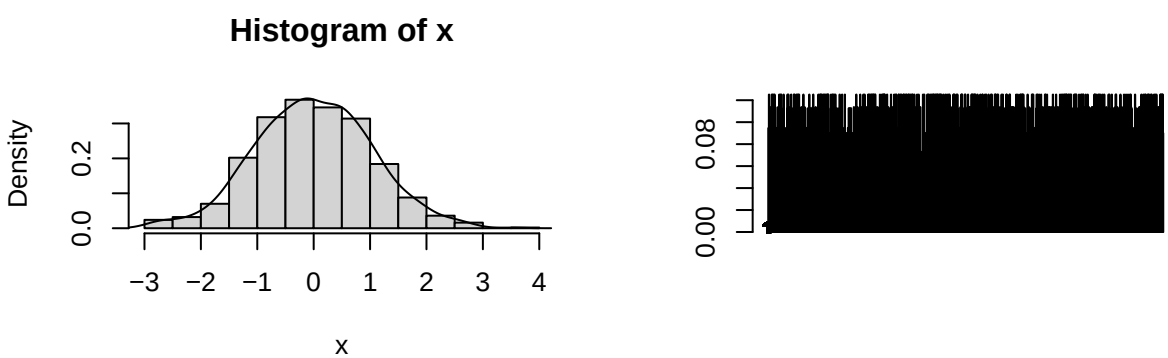

```
boxplot(marks~age,data=df)
```




```
#Adding multiple plots in the same frame
par(mfrow=c(2,2))
x=rnorm(100)
y=rpois(100,10)
plot(ecdf(x))
plot(dpois(y,10))
```

```r
boxplot(rnorm(100))
plot(rnorm(10))
```




```r
#Overlaying plots
x=rnorm(1000)
hist(x,probability=TRUE)
lines(density(x))
y=rpois(1000,10)
z=dpois(y,10)
barplot(z)
pt=prop.table(table(x))
points(x,pt,pch="19")
```